“数据分析与实践”实验二 实验报告

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实验目的

实验在python环境下，利用Python标准库以及Requests库中的所有方法，获取Nature平台上关心的论文信息，需要实现的任务包括：

1. (20%) 进入Nature主页，通过局级检索功能，搜索关键词llm, 限制年份为“2023-2024"，搜索得到近两年间关键词含有llm的Nature平台相关文章。

   使用python代码获取检索结果第一页的html内容并解码为文本串，将其以UTF-8编码格式写入page1.txt文件中，用于后续处理。
2. (20%) 打开page1.txt, 观察相关数据的组织格式规律。从这些文本串中提取一些论文相关的重要信息(包括文章标题，文章地址，文章简介，作者列表，文章类型，期刊名称，卷宗/页面信息)，并按照发表的期刊名进行分类，以指定格式存储到一个字典列表中。

   基于上述结果，输出每个期刊在page1中包含的论文数量。
3. (30%) 某篇文章打开后的界面中，观察可以发现该文章的作者信息实际上多于搜索结果中展示的内容，仔细观察此界面的html数据组织格式，依据此编写python程序，将上一步骤中的字典提取内容中的作者列表中的内容进行替换，替换为文章主页面显示的全部作者。
4. (20%）JSON是一种轻量级的数据交互格式，常用于存储和表示数据。

   将上一步获得的字典列表转化为 json 对象（可使用json包的相应方法），并以2字符缩进的方式写入nature_llm.json文件中。
5. （选做）Nature平台的Advanced Reasearch页面中，还可以依据期刊名称和Volume进行检索。

   对于上述提取到的pageI中的文章，请你尝试提取所有发表在以Nature为开头的期刊（不包括Nature Reviews开头的期刊）的文章，请你检索得到这些文章所在的相应Volume中的所有文章信息，提取检索结果的pageI中的所有文章，并以如下格式存储在volume.json文件中，重复的Volume只需要提取一次。查找过程需要使用python代码自动实现，不可以手动获取网址。 （此处的author信息只使用检索界面的部分信息即可）

程序设计

一、获取page1信息

在浏览器中搜索获得page1的url为 https://www.nature.com/search?q=llm&order=relevance&date_range=2023-2024。通过Requests库的get方法获取网页的html内容并解码为文本串，通过文件操作写入page1.txt中。

```
import requests

url = 'https://www.nature.com/search?q=llm&order=relevance&date_range=2023-2024'
response = requests.get(url)
#获取page1的html

if response.status_code == 200:
    data = response.text
else:
```

```
    print('Request failed with status code:',response.status_code)
#解码为字符串data

with open('page1.txt','w', encoding = 'UTF_8') as file1:
    file1.write(data)
#写入文件page1.txt
```

二、分析page1信息

逐行读入page1.txt（在任务5中会使用整体处理的方法）进行分析。

通过观察发现html代码中每篇文章的内容均单独存在一个article块中，使用标记变量标记article块的开始与结束，同时根据信息的编码格式，通过正则表达式捕捉。

捕捉完单篇文章的所有信息后，根据其期刊，存入列表paperlist。

```
import re

paperlist = []

with open('page1.txt','r', encoding = 'UTF_8') as file1:
    start = 0
    #标记article块是否开始
    end = 0
    #标记article块是否结束
    authors = []
    description = ''
    for line in file1:
        if not start:
            start = re.search('<article',line)
            #匹配article块开头
        else:
            end = re.search('</article>',line)
            #匹配article块结尾
            if end:
                if description == '' :
                    description = 'no description'
                #处理没有descreption的情况
                flag = 0
                #标记期刊是否存在
                for dict1 in paperlist:
                    if dict1["journal"] == journal:
                        (dict1["papers"]).append({"title": title, "authors":
authors, "url": paperurl, "description": description, "type": type,
"volume_page_info": page})
                        flag = 1
                        break
                if flag == 0:
                    paperlist.append({"journal": journal, "papers": [{"title":
title, "authors": authors, "url": paperurl, "description": description, "type":
type, "volume_page_info": page}]})
                start = 0
                end = 0
                authors = []
                description = ''
                #找到结尾，存储所有信息到paperlist，重置start,end,authors和
description
```

```
                else:
                    matcht = re.search('data-track-label="link">(.*)</a>',line)
                    if matcht:
                        title = matcht.group(1)
                    #匹配论文标题
                    matchdes = re.search('<p>(.*)</p>',line)
                    if matchdes:
                        description = matchdes.group(1)
                    #匹配论文描述
                    matchurl = re.search('<a href="(.*)"',line)
                    if matchurl:
                        paperurl = matchurl.group(1)
                    #匹配论文url
                    authors += re.findall('<span itemprop="name">(.*?)</span>',line)
                    #匹配论文作者
                    matchtype = re.search('<span class="c-meta__type">(.*?)
</span>',line)
                    if matchtype:
                        type = matchtype.group(1)
                    #匹配论文类型
                    matchjournal = re.search('data-test="journal-title-and-link">
(.*?)</div>',line)
                    if matchjournal:
                        journal = matchjournal.group(1)
                    #匹配论文类型
                    matchpage = re.search('data-test="volume-and-page-info">(.*?)
</div>',line)
                    if matchpage:
                        page = matchpage.group(1)
                    #匹配论文类型
#完成信息存储
```

收集完毕后，输出每个期刊在page1中包含的论文数量。

```
for dict1 in paperlist:
    print(f'期刊名：{dict1['journal']}，论文数量：{len(dict1['papers'])}')
#输出每个期刊在page1中包含的论文数量
```

三、进一步获取文章内容

在任务2获取的信息列表paperlist中包括了论文详情页url的后缀"url"。再次通过get方法获取详情页的html并解码为文本串。

```
for dict1 in paperlist:
    for paper1 in dict1["papers"]:
        urlp = paper1["url"]
        url = 'https://www.nature.com' + urlp
        #取出上述论文信息中的url后缀，加上前缀得到详情页的url
        response = requests.get(url)
        if response.status_code == 200:
            data = response.text
        else:
            print('Request failed with status code:',response.status_code)
        #得到论文详情页的html代码
```

再次进行作者信息匹配，注意到作者在详情页中是以列表形式表示的，可以直接用eval函数完成转换。

```
        match = re.search('"authors":(.*?),"publishedAt"',data)
        #匹配作者
        if match:
            authors = eval(match.group(1))
            #作者在html编码格式中为list形式，直接读入列表表达式并用eval函数转换即可
            paper1["authors"] = authors
            #更新作者信息
```

任务5中需要文章所在的期刊代码，在文章详情页中给出，故为节省时间和减少访问次数，在这里一并收集，具体情况在任务5中讨论。

四、存储为json文件

使用json包的dump方法即可方便地将paperlist存储为json文件。

```
import json

with open('nature_llm.json','w',encoding = 'UTF_8') as fj:
    json.dump(paperlist, fj, indent=2)
    #以2字符缩进将列表存储到nature_llm.json文件中
```

五、获取期刊内容

首先需要在文章详情页中获取期刊的代码，在任务3中完成：

```
journallist = []
# 存储期刊代码与期数，用于任务5

# 在获取作者信息后：
        matchjournal = re.search('"journal":{"pcode":"(.*?)","title":"([Nn]ature(?! R[r]eviews).*?)","volume"',data)
        matchvolume = re.search('"volume":"(.*?)","issue"',data)
        #为任务5作准备，匹配期刊代码、期数与期刊名，期刊名以Nature/nature开头且不以Nature Reviews开头
        #实际上Nature Reviews开头的期刊html内会因为volume格式为"volume":null而不被匹配到
        if matchjournal and matchvolume:
            journal = matchjournal.group(1)
            volume = matchvolume.group(1)
            journalname = matchjournal.group(2)
            if (journal, volume, journalname) not in journallist:
                journallist.append((journal, volume, journalname))
```

期刊内容的url为 "https://www.nature.com/search?volume={volume}&order=relevance&journal={journal}"。将{volume}替换为期数，{journal}替换为期刊代码即可。

```
volumelist = []

for journal, volume, journalname in journallist:
    newdict = {"journal": journalname, "volume": volume, "papers": []}

    url = f'https://www.nature.com/search?volume=
{volume}&order=relevance&journal={journal}'
    response = requests.get(url)
    if response.status_code == 200:
        data = response.text
    else:
        print('Request failed with status code:',response.status_code)
    #获取网页html
```

接下来用re库的compile方法和findall函数获取所有article块，存储在matches列表中。

```
    pattern = re.compile(r'<article\s+class[^>]*>(.*?)</article>', re.DOTALL)
    matches = re.findall(pattern,data)
    #找到所有article块
```

接下来用和任务2大体类似的方法获取论文信息，并一并存储在volumelist中。

```
    authors = []
    description = ''
    for article in matches:
        matcht = re.search('data-track-label="link">(.*)</a>',article)
        if matcht:
            title = matcht.group(1)
        #匹配论文标题
        matchdes = re.search('<p>(.*)</p>',article)
        if matchdes:
            description = matchdes.group(1)
        #匹配论文描述
        matchurl = re.search('<a href="(.*)"',article)
        if matchurl:
            paperurl = matchurl.group(1)
        #匹配论文url
        authors += re.findall('<span itemprop="name">(.*?)</span>',article)
        #匹配论文作者
        matchtype = re.search('<span class="c-meta__type">(.*?)</span>',article)
        if matchtype:
            type = matchtype.group(1)
        #匹配论文类型

        if description == '' :
            description = 'no description'
        #处理没有descreption的情况

        (newdict["papers"]).append({"title": title, "authors": authors, "url":
paperurl, "description": description, "type": type})
        #更新newdict内papers的内容

        authors = []
        description = ''
        #重置authors和description
```

```
    volumelist.append(newdict)
```

最后用json库的dump方法存储为json文件。

```
with open('volume.json','w',encoding = 'UTF_8') as fj:
    json.dump(volumelist, fj, indent=2)
    #以2字符缩进将列表存储到volume.json文件中
#完成信息存储
```

测试结果

一、获取page1信息

正常运行，用时2.1s，page1.txt内容见附录。

二、分析page1信息

正常运行，用时0.0s，输出：

```
期刊名：Nature Machine Intelligence，论文数量：6
期刊名：Nature Communications，论文数量：10
期刊名：Scientific Reports，论文数量：12
期刊名：Nature Methods，论文数量：1
期刊名：npj Digital Medicine，论文数量：3
期刊名：Nature Medicine，论文数量：2
期刊名：Nature，论文数量：4
期刊名：Nature Human Behaviour，论文数量：2
期刊名：npj Precision Oncology，论文数量：1
期刊名：BDJ Open，论文数量：1
期刊名：Nature Computational Science，论文数量：2
期刊名：Nature Reviews Urology，论文数量：1
期刊名：Communications Materials，论文数量：1
期刊名：npj Biodiversity，论文数量：1
期刊名：Humanities and Social Sciences Communications，论文数量：1
期刊名：Eye，论文数量：1
期刊名：npj Computational Materials，论文数量：1
```

三、进一步获取文章内容

正常运行，用时2min14.3s。

四、存储为json文件

正常运行，用时0.1s，nature_llm.json内容见附录。

五、获取期刊内容

正常运行，用时26.8s，volume.json内容见附录。

附录

page1.txt

```
<!DOCTYPE html>
<html lang="en" class="grade-c">
<head>
    <title>llm | Nature Search Results</title>

<link rel="preconnect" href="https://cmp.nature.com" crossorigin>

<meta http-equiv="X-UA-Compatible" content="IE=edge">
<meta name="applicable-device" content="pc,mobile">
<meta name="viewport" content="width=device-width,initial-scale=1.0,maximum-
scale=5,user-scalable=yes">

<script data-test="dataLayer">
    window.dataLayer = [{"content":{"category":
{"contentType":null,"legacy":null},"article":null,"attributes":null,"contentInfo
":null,"journal":null,"authorization":{"status":true},"features":
[],"collection":null},"page":{"category":
{"pageType":"search_results"},"attributes":{"template":"mosaic","featureFlags":
[{"name":"nature-onwards-journey","active":false}],"testGroup":null},"search":
{"keywords":"llm","articleType":null,"journal":null,"subject":null,"dateRange":n
ull,"sortOrder":"Relevance"}},"privacy":{},"version":"1.0.0","product":
{"jobPost":null},"session":null,"user":null,"backHalfContent":false,"country":"C
N","hasBody":false,"uneditedManuscript":false,"twitterId":
["o3xnx","o43y9","o3ef7"],"baiduId":"d38bce82bcb44717ccc29a90c4b781ea","japan":f
alse}];
    window.dataLayer.push({
        ga4MeasurementId: 'G-ERRNTNZ807',
        ga360TrackingId: 'UA-71668177-1',
        twitterId: ['3xnx', 'o43y9', 'o3ef7'],
        baiduId: 'd38bce82bcb44717ccc29a90c4b781ea',
        ga4ServerUrl: 'https://collect.nature.com',
        imprint: 'nature'
    });
</script>

<script>
    (function(w, d) {
        w.config = w.config || {};
        w.config.mustardcut = false;


        if (w.matchMedia && w.matchMedia('only print, only all and (prefers-
color-scheme: no-preference), only all and (prefers-color-scheme: light), only
all and (prefers-color-scheme: dark)').matches) {
            w.config.mustardcut = true;
            d.classList.add('js');
            d.classList.remove('grade-c');
            d.classList.remove('no-js');
        }
    })(window, document.documentElement);
</script>




    <link href="/static/css/grade-c-c8a47957e2.css" rel="stylesheet"/>
```

```
    <link data-test="custom-css" href="/static/css/search/search-150-
fdfe762873.css" rel="stylesheet"  media="only print, only all and (prefers-
color-scheme: no-preference), only all and (prefers-color-scheme: light), only
all and (prefers-color-scheme: dark)" />




    <style>
        .c-header {
            border-bottom: 5px solid #d9d9d9;
        }
    </style>

<link rel="apple-touch-icon" sizes="180x180"
href=/static/images/favicons/nature/apple-touch-icon-f39cb19454.png>
<link rel="icon" type="image/png" sizes="48x48"
href=/static/images/favicons/nature/favicon-48x48-b52890008c.png>
<link rel="icon" type="image/png" sizes="32x32"
href=/static/images/favicons/nature/favicon-32x32-3fe59ece92.png>
<link rel="icon" type="image/png" sizes="16x16"
href=/static/images/favicons/nature/favicon-16x16-951651ab72.png>
<link rel="manifest" href=/static/manifest.json crossorigin="use-credentials">
<link rel="mask-icon" href=/static/images/favicons/nature/safari-pinned-tab-
69bff48fe6.svg color="#000000">
<link rel="shortcut icon" href=/static/images/favicons/nature/favicon.ico>
<meta name="msapplication-TileColor" content="#000000">
<meta name="msapplication-config" content=/static/browserconfig.xml>
<meta name="theme-color" content="#000000">
<meta name="application-name" content="Nature">


<script>
    (function () {
        if ( typeof window.CustomEvent === "function" ) return false;
        function CustomEvent ( event, params ) {
            params = params || { bubbles: false, cancelable: false, detail: null
};
            var evt = document.createEvent( 'CustomEvent' );
            evt.initCustomEvent( event, params.bubbles, params.cancelable,
params.detail );
            return evt;
        }

        CustomEvent.prototype = window.Event.prototype;

        window.CustomEvent = CustomEvent;
    })();
</script>



<!-- Google Tag Manager -->
<script data-test="gtm-head">
    window.initGTM = function() {
        if (window.config.mustardcut) {
            (function (w, d, s, l, i) {
                w[l] = w[l] || [];
```

```
                w[l].push({'gtm.start': new Date().getTime(), event: 'gtm.js'});
                var f = d.getElementsByTagName(s)[0],
                        j = d.createElement(s),
                        dl = l != 'dataLayer' ? '&l=' + l : '';
                j.async = true;
                j.src = 'https://www.googletagmanager.com/gtm.js?id=' + i + dl;
                f.parentNode.insertBefore(j, f);
            })(window, document, 'script', 'dataLayer', 'GTM-MRVXSHQ');
        }
    }
</script>
<!-- End Google Tag Manager -->

    <script>
    (function(w,d,t) {
        function cc() {
            var h = w.location.hostname;
            if (h.indexOf('preview-www.nature.com') > -1) return;

            var e = d.createElement(t),
                    s = d.getElementsByTagName(t)[0];

            if (h.indexOf('nature.com') > -1) {
                if (h.indexOf('test-www.nature.com') > -1) {
                    e.src = 'https://cmp.nature.com/production_live/en/consent-
bundle-8-77.js';
                    e.setAttribute('onload',
"initGTM(window,document,'script','dataLayer','GTM-MRVXSHQ')");
                } else {
                    e.src = 'https://cmp.nature.com/production_live/en/consent-
bundle-8-77.js';
                    e.setAttribute('onload',
"initGTM(window,document,'script','dataLayer','GTM-MRVXSHQ')");
                }
            } else {
                e.src = '/static/js/cookie-consent-es5-bundle-cb57c2c98a.js';
                e.setAttribute('data-consent', h);
            }
            s.insertAdjacentElement('afterend', e);
        }

        cc();
    })(window,document,'script');
</script>


<script id="js-position0">
    (function(w, d) {
        w.idpVerifyPrefix = 'https://verify.nature.com';
        w.ra21Host = 'https://wayf.springernature.com';
        var moduleSupport = (function() {
            return 'noModule' in d.createElement('script');
        })();

        if (w.config.mustardcut === true) {
            w.loader = {
                index: 0,
                registered: [],
```

```
                scripts: [

                        {src: '/static/js/search-150-es5-bundle-145c24d967.js', 
test: 'alternate-path-js'},
                        {src: '/static/js/header-150-es6-bundle-5bb959eaa1.js', 
test: 'header-150-js', module: true},
                        {src: '/static/js/header-150-es5-bundle-994fde5b1d.js', 
test: 'header-150-js', nomodule: true}

                ].filter(function (s) {
                    if (s.src === null) return false;
                    if (moduleSupport && s.nomodule) return false;
                    return !(!moduleSupport && s.module);
                }),

                register: function (value) {
                    this.registered.push(value);
                },

                ready: function () {
                    if (this.registered.length === this.scripts.length) {
                        this.registered.forEach(function (fn) {
                            if (typeof fn === 'function') {
                                setTimeout(fn, 0);
                            }
                        });
                        this.ready = function () {};
                    }
                },

                insert: function (s) {
                    var t = d.getElementById('js-position' + this.index);
                    if (t && t.insertAdjacentElement) {
                        t.insertAdjacentElement('afterend', s);
                    } else {
                        d.head.appendChild(s);
                    }
                    ++this.index;
                },

                createScript: function (script, beforeLoad) {
                    var s = d.createElement('script');
                    s.id = 'js-position' + (this.index + 1);
                    s.setAttribute('data-test', script.test);
                    if (beforeLoad) {
                        s.defer = 'defer';
                        s.onload = function () {
                            if (script.noinit) {
                                loader.register(true);
                            }
                            if (d.readyState === 'interactive' || d.readyState 
=== 'complete') {
                                loader.ready();
                            }
                        };
                    } else {
                        s.async = 'async';
                    }
```

```
                    s.src = script.src;
                    return s;
                },

                init: function () {
                    this.scripts.forEach(function (s) {
                        loader.insert(loader.createScript(s, true));
                    });

                    d.addEventListener('DOMContentLoaded', function () {
                        loader.ready();
                        var conditionalScripts;

                            conditionalScripts = [
                                {match: 'math,span.mathjax-tex', src:
'/static/js/math-es6-bundle-23597ae350.js', test: 'math-js', module: true},
                                {match: 'math,span.mathjax-tex', src:
'/static/js/math-es5-bundle-6532c6f78b.js', test: 'math-js', nomodule: true}
                            ];


                        if (conditionalScripts) {
                            conditionalScripts.filter(function (script) {
                                return !!document.querySelector(script.match) &&
!((moduleSupport && script.nomodule) || (!moduleSupport && script.module));
                            }).forEach(function (script) {
                                loader.insert(loader.createScript(script));
                            });
                        }
                    }, false);
                }
            };
            loader.init();
        }
    })(window, document);
</script>
```

```
<meta name="robots" content="noindex,follow">
<meta name="access" content="Yes">

<link rel="search" href="https://www.nature.com/search">
<link rel="search" href="https://www.nature.com/opensearch/opensearch.xml"
type="application/opensearchdescription+xml" title="nature.com">
<link rel="search" href="https://www.nature.com/opensearch/request"
type="application/sru+xml" title="nature.com">
```

```
</head>
<body>

<noscript><iframe src="https://www.googletagmanager.com/ns.html?id=GTM-MRVXSHQ"
                  height="0" width="0" style="display:none;visibility:hidden">
</iframe></noscript>





<div class="position-relative cleared z-index-50 background-white" data-
test="top-containers">
    <a class="c-skip-link" href="#content">Skip to main content</a>



<div class="c-grade-c-banner u-hide">
    <div class="c-grade-c-banner__container">

        <p>Thank you for visiting nature.com. You are using a browser version
with limited support for CSS. To obtain
            the best experience, we recommend you use a more up to date browser
(or turn off compatibility mode in
            Internet Explorer). In the meantime, to ensure continued support, we
are displaying the site without styles
            and JavaScript.</p>

    </div>
</div>







        <div class="u-hide u-show-following-ad"></div>

    <aside class="c-ad c-ad--728x90">
        <div class="c-ad__inner" data-container-type="banner-advert">
            <p class="c-ad__label">Advertisement</p>



    <div id="div-gpt-ad-top-1"
         class="div-gpt-ad advert leaderboard js-ad text-center hide-print
grade-c-hide"
         data-ad-type="leaderboard"
         data-pa11y-ignore
         data-test="top-ad"
         data-gpt
         data-gpt-unitpath="/285/nature.com/search_results"
         data-gpt-sizes="728x90"
```

```
         data-gpt-
targeting="pos&#x3D;top;type&#x3D;search_results;path&#x3D;/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024;kw&#x3D;llm">

        <noscript>
            <a href="//pubads.g.doubleclick.net/gampad/jump?
iu=/285/nature.com/search_results&amp;sz=728x90&amp;pos&#x3D;top;type&#x3D;searc
h_results;path&#x3D;/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024;kw&#x3D;llm">
                <img data-test="gpt-advert-fallback-img"
                     src="//pubads.g.doubleclick.net/gampad/ad?
iu=/285/nature.com/search_results&amp;sz=728x90&amp;pos&#x3D;top;type&#x3D;searc
h_results;path&#x3D;/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024;kw&#x3D;llm"
                     alt="Advertisement"
                     width="728"
                     height="90"/>
            </a>
        </noscript>
    </div>




        </div>
    </aside>



    <header class="c-header" id="header" data-header data-track-
component="nature-150-split-header">
        <div class="c-header__row">
            <div class="c-header__container">
                <div class="c-header__split">



                    <div class="c-header__logo-container">

                        <a href="/"
                           data-track="click" data-track-action="home" data-
track-label="image">
                            <picture class="c-header__logo">
                                <source srcset="/static/images/product-
logos/naturecom-primary-5cd474411f.svg" media="(min-width: 875px)">
                                <img src="/static/images/product-
logos/naturecom-secondary-5cd474411f.svg" height="32" alt="Nature">
                            </picture>
                        </a>

                    </div>

                    <ul class="c-header__menu c-header__menu--global">
                        <li class="c-header__item c-header__item--padding c-
header__item--hide-md-max">
                            <a class="c-header__link"
href="https://www.nature.com/siteindex" data-test="siteindex-link"
                               data-track="click" data-track-action="open nature
research index" data-track-label="link">
                                <span>View all journals</span>
                            </a>
```

```
                        </li>
                        <li class="c-header__item c-header__item--padding c-
header__item--pipe">
                            <a class="c-header__link c-header__link--search"
                                href="#search-menu"
                                data-header-expander
                                data-test="search-link" data-track="click" data-
track-action="open search tray" data-track-label="button">
                                <svg role="img" aria-hidden="true"
focusable="false" height="22" width="22" viewBox="0 0 18 18"
xmlns="http://www.w3.org/2000/svg"><path d="M16.48 15.455c.283.282.29.749.007
1.032a.738.738 0 01-1.032-.007l-3.045-3.044a7 7 0 111.026-1.026zM8 14A6 6 0 108
2a6 6 0 000 12z"/></svg><span>Search</span>
                            </a>
                        </li>
                        <li class="c-header__item c-header__item--padding c-
header__item--snid-account-widget c-header__item--pipe">

                                <a class="c-header__link eds-c-header__link"
id="identity-account-widget" data-track="click_login" data-track-
context="header" href='https://idp.nature.com/auth/personal/springernature?
redirect_uri=https://www.nature.com/search?
q=llm&order=relevance&date_range=2023-2024'><span class="eds-c-header__widget-
fragment-title">Log in</span></a>

                        </li>
                    </ul>
                </div>
            </div>
        </div>

    </header>




        <nav class="u-mb-16" aria-label="breadcrumbs">
            <div class="u-container">
                <ol class="c-breadcrumbs" itemscope
itemtype="https://schema.org/BreadcrumbList">
                    <li class="c-breadcrumbs__item" id="breadcrumb0"
itemprop="itemListElement" itemscope itemtype="https://schema.org/ListItem"><a
class="c-breadcrumbs__link"
                               href="/" itemprop="item"
                               data-track="click" data-track-action="breadcrumb"
data-track-category="header" data-track-label="link:nature"><span
itemprop="name">nature</span></a><meta itemprop="position" content="1">
                                    <svg class="c-breadcrumbs__chevron"
role="img" aria-hidden="true" focusable="false" height="10" viewBox="0 0 10 10"
width="10"
                                         xmlns="http://www.w3.org/2000/svg">
```

```
                                        <path d="m5.96738168 4.70639573
2.39518594-2.41447274c.37913917-.38219212.98637524-.38972225
1.35419292-.01894278.37750606.38054586.37784436.99719163-.00013556 1.37821513l-
4.03074001 4.06319683c-.37758093.38062133-.98937525.38100976-
1.367372-.00003075l-4.03091981-
4.06337806c-.37759778-.38063832-.38381821-.99150444-.01600053-
1.3622839.37750607-.38054587.98772445-.38240057 1.37006824.00302197l2.39538588
2.4146743.96295325.98624457z"
                                              fill="#666" fill-rule="evenodd"
transform="matrix(0 -1 1 0 0 10)"/>
                                    </svg>
                                </li><li class="c-breadcrumbs__item"
id="breadcrumb1" itemprop="itemListElement" itemscope
itemtype="https://schema.org/ListItem">
                                    <span itemprop="name">search</span><meta
itemprop="position" content="2"></li>
                </ol>
            </div>
        </nav>









</div>



<div id="content">

    <div class="u-container u-mb-16" data-test="search-results-title"
role="main">
        <search role="search" aria-label="search from body">
            <form action="/search/submit"
                  method="post"
                  autocomplete="off"
                  data-component="search-form"
                  data-featureflags_persist_filters="true">
                <div class="c-search c-search--max-width u-mb-24">
                    <h1 class="u-ma-0"><label for="search-keywords" class="u-
display-inline-block u-mb-16">Search</label></h1>
                    <div class="c-search__field">
                        <div class="c-search__input-container c-search__input-
container--sm">
                            <input type="search"
                                   id="search-keywords"
                                   data-test="search-box"
                                   class="c-search__input"
                                   name="q"
                                   value="llm"
                                   >
                        </div>
                        <div class="c-search__button-container">
                            <button type="submit" data-test="search-button"
class="c-search__button" data-action="clear-adv">
                                <span class="c-search__button-
text">Search</span>
```

```
                                <svg class="u-flex-static" role="img" aria-
hidden="true" focusable="false" height="16"
                                     width="16" viewBox="0 0 16 16"
xmlns="http://www.w3.org/2000/svg">
                                    <path d="M16.48 15.455c.283.282.29.749.007
1.032a.738.738 0 01-1.032-.007l-3.045-3.044a7 7 0 111.026-1.026zM8 14A6 6 0 108
2a6 6 0 000 12z"></path>
                                </svg>
                            </button>
                        </div>
                        <a href="/search/advanced?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024" class="c-
search__link"
                           data-track="click" data-track-category="search
results page n150"
                           data-track-action="click link to advanced search
page"
                           data-track-label="(not set)"
                           data-test="advance-search-link" tabindex="0">Advanced
search</a>
                    </div>

                </div>



                        <div class="c-facet c-facet--small u-mb-16" data-facet
data-test="search-filter-box">
                            <h2 class="u-visually-hidden">Filter By:</h2>
                            <div class="c-facet__container">

    <fieldset class="c-facet__item u-pa-0" data-test="field-set-journals">
        <legend>
            <span class="c-facet__label" id="journal-legend">Journal</span>
            <span class="grade-c-hide u-hide u-js-show">
            <button  type="button" class="c-facet__button c-facet__button--
border "
                    aria-labelledby="journal-legend"
                    data-facet-expander
                    data-facet-target="#journal-target"
                    data-track="click" data-track-action="journal filter" data-
track-label="button">
                <svg class="c-facet__icon" role="img" aria-hidden="true"
focusable="false" height="16"
                     viewBox="0 0 16 16" width="16"
xmlns="http://www.w3.org/2000/svg"><path
                        d="m5.58578644 3-3.29289322-
3.29289322c-.39052429-.39052429-.39052429-1.02368927 0-
1.41421356s1.02368927-.39052429 1.41421356 0l4 4c.39052429.39052429.39052429
1.02368927 0 1.41421356l-4 4c-.39052429.39052429-1.02368927.39052429-1.41421356
0s-.39052429-1.02368927 0-1.41421356z"
                        transform="matrix(0 1 -1 0 11 3)"></path></svg>
            </button>
        </span>
        </legend>
        <span class="u-visually-hidden">Check one or more journals to show
results from those journals only.</span>
        <div id="journal-target" data-test="journal-target" data-facet-
analytics="journal" class="c-facet-expander u-js-hide">
```

```
            <ul class="c-facet-expander__list">

                    <li class="c-facet-expander__list-item">
                        <input name="journal" id="journal-srep" value="srep"
                               data-action="submit"
                               type="checkbox" />
                        <label class="c-facet-expander__link" for="journal-
srep">
                            <span>Scientific Reports (132)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="journal" id="journal-nature" value="nature"
                               data-action="submit"
                               type="checkbox" />
                        <label class="c-facet-expander__link" for="journal-
nature">
                            <span>Nature (127)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="journal" id="journal-npjdigitalmed"
value="npjdigitalmed"
                               data-action="submit"
                               type="checkbox" />
                        <label class="c-facet-expander__link" for="journal-
npjdigitalmed">
                            <span>npj Digital Medicine (56)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="journal" id="journal-ncomms" value="ncomms"
                               data-action="submit"
                               type="checkbox" />
                        <label class="c-facet-expander__link" for="journal-
ncomms">
                            <span>Nature Communications (42)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="journal" id="journal-natmachintell"
value="natmachintell"
                               data-action="submit"
                               type="checkbox" />
                        <label class="c-facet-expander__link" for="journal-
natmachintell">
                            <span>Nature Machine Intelligence (34)</span>
                        </label>
                    </li>

            </ul>
            <p class="u-mb-0">
```

```
                <a href="/search/advanced?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024#journals"
data-track="click"
                   data-track-category="search results page n150" data-track-
action="click choose more link in journal filter"
                   data-track-label="(not set)" class="c-facet-expander__link c-
facet-expander__link--underline"
                   data-test="advance-search-link-journals">Choose more<span
class="u-visually-hidden"> journals</span>
                </a>
            </p>
        </div>
    </fieldset>



    <fieldset class="c-facet__item u-pa-0" data-test="field-set-article-type">
        <legend>
            <span class="c-facet__label" id="article-legend">Article type</span>
            <span class="grade-c-hide u-hide u-js-show">
                <button type="button"  class="c-facet__button c-facet__button--
border "
                        aria-labelledby="article-legend"
                        data-facet-expander data-facet-target="#article-type-
target"
                        data-track="click" data-track-action="article type
filter" data-track-label="button">
                    <svg class="c-facet__icon" role="img" aria-hidden="true"
focusable="false" height="16" viewBox="0 0 16 16" width="16"
xmlns="http://www.w3.org/2000/svg"><path d="m5.58578644 3-3.29289322-
3.29289322c-.39052429-.39052429-.39052429-1.02368927 0-
1.41421356s1.02368927-.39052429 1.41421356 0l4 4c.39052429.39052429.39052429
1.02368927 0 1.41421356l-4 4c-.39052429.39052429-1.02368927.39052429-1.41421356
0s-.39052429-1.02368927 0-1.41421356z" transform="matrix(0 1 -1 0 11 3)"></path>
</svg>
                </button>
            </span>
        </legend>
        <span class="u-visually-hidden">Check one or more article types to show
results from those article types only.</span>
        <div id="article-type-target" data-test="article-type-target" data-
facet-analytics="article type" class="c-facet-expander u-js-hide">
            <ul class="c-facet-expander__list">

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-research"
value="research"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
research">
                            <span>Research (354)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-news"
value="news"
```

```
                                type="checkbox"
                                />
                        <label class="c-facet-expander__link" for="article-type-
news">
                            <span>News (77)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-comments-
and-opinion" value="comments-and-opinion"
                                type="checkbox"
                                />
                        <label class="c-facet-expander__link" for="article-type-
comments-and-opinion">
                            <span>Comments &amp; Opinion (75)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-reviews"
value="reviews"
                                type="checkbox"
                                />
                        <label class="c-facet-expander__link" for="article-type-
reviews">
                            <span>Reviews (62)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-editorial"
value="editorial"
                                type="checkbox"
                                />
                        <label class="c-facet-expander__link" for="article-type-
editorial">
                            <span>Editorial (23)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-
correspondence" value="correspondence"
                                type="checkbox"
                                />
                        <label class="c-facet-expander__link" for="article-type-
correspondence">
                            <span>Correspondence (22)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-news-and-
views" value="news-and-views"
                                type="checkbox"
                                />
```

```
                        <label class="c-facet-expander__link" for="article-type-
news-and-views">
                            <span>News &amp; Views (19)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-research-
highlights" value="research-highlights"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
research-highlights">
                            <span>Research Highlights (18)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-special-
features" value="special-features"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
special-features">
                            <span>Special Features (9)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-advertorial"
value="advertorial"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
advertorial">
                            <span>Advertorial (1)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-amendments-
and-corrections" value="amendments-and-corrections"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
amendments-and-corrections">
                            <span>Amendments and Corrections (1)</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="article_type" id="article-type-books-and-
arts" value="books-and-arts"
                               type="checkbox"
                               />
                        <label class="c-facet-expander__link" for="article-type-
books-and-arts">
                            <span>Books &amp; Arts (1)</span>
```

```
                        </label>
                    </li>

                </ul>
            </div>
        </fieldset>


        <fieldset class="c-facet__item u-pa-0" data-test="field-set-subjects">
            <legend>
                <span class="c-facet__label" id="subject-legend">Subject</span>
                <span class="grade-c-hide u-hide u-js-show">
                <button type="button"  class="c-facet__button c-facet__button--
border "
                        aria-labelledby="subject-legend"
                        data-facet-expander
                        data-facet-target="#subject-target"
                        data-track="click" data-track-action="subject filter" data-
track-label="button">
                    <svg class="c-facet__icon" role="img" aria-hidden="true"
focusable="false" height="16"
                         viewBox="0 0 16 16" width="16"
xmlns="http://www.w3.org/2000/svg"><path
                            d="m5.58578644 3-3.29289322-
3.29289322c-.39052429-.39052429-.39052429-1.02368927 0-
1.41421356s1.02368927-.39052429 1.41421356 0l4 4c.39052429.39052429.39052429
1.02368927 0 1.41421356l-4 4c-.39052429.39052429-1.02368927.39052429-1.41421356
0s-.39052429-1.02368927 0-1.41421356z"
                            transform="matrix(0 1 -1 0 11 3)"></path></svg>
                </button>
            </span>
            </legend>
            <span class="u-visually-hidden">Check one or more subjects to show
results from those subjects only.</span>
            <div id="subject-target" data-test="subject-target" data-facet-
analytics="subject" class="c-facet-expander u-js-hide">
                <ul class="c-facet-expander__list">

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
biological-techniques" value="biological-techniques"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
biological-techniques">
                            <span>Biological techniques</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
business-and-industry" value="business-and-industry"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
business-and-industry">
                            <span>Business and industry</span>
                        </label>
                    </li>
```

```
                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
cancer" value="cancer"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
cancer">
                            <span>Cancer</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
chemistry" value="chemistry"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
chemistry">
                            <span>Chemistry</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
computational-biology-and-bioinformatics" value="computational-biology-and-
bioinformatics"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
computational-biology-and-bioinformatics">
                            <span>Computational biology and
bioinformatics</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
diseases" value="diseases"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
diseases">
                            <span>Diseases</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-drug-
discovery" value="drug-discovery"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-drug-
discovery">
                            <span>Drug discovery</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
engineering" value="engineering"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
engineering">
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                            <span>Engineering</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
environmental-social-sciences" value="environmental-social-sciences"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
environmental-social-sciences">
                            <span>Environmental social sciences</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
genetics" value="genetics"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
genetics">
                            <span>Genetics</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
health-care" value="health-care"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
health-care">
                            <span>Health care</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
language-and-linguistics" value="language-and-linguistics"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
language-and-linguistics">
                            <span>Language and linguistics</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
materials-science" value="materials-science"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
materials-science">
                            <span>Materials science</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
mathematics-and-computing" value="mathematics-and-computing"
                               data-action="submit"/>
```

```
                        <label class="c-facet-expander__link" for="subject-
mathematics-and-computing">
                            <span>Mathematics and computing</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
medical-research" value="medical-research"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
medical-research">
                            <span>Medical research</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
neuroscience" value="neuroscience"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
neuroscience">
                            <span>Neuroscience</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
psychology" value="psychology"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
psychology">
                            <span>Psychology</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
science-technology-and-society" value="science-technology-and-society"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
science-technology-and-society">
                            <span>Science, technology and society</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input type="checkbox" name="subject" id="subject-
scientific-community" value="scientific-community"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
scientific-community">
                            <span>Scientific community</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
```

```
                        <input type="checkbox" name="subject" id="subject-
social-sciences" value="social-sciences"
                               data-action="submit"/>
                        <label class="c-facet-expander__link" for="subject-
social-sciences">
                            <span>Social sciences</span>
                        </label>
                    </li>

            </ul>
        </div>
    </fieldset>



    <fieldset class="c-facet__item u-pa-0">
        <legend>
            <span class="c-facet__label" id="date-legend">Date</span>
            <span class="grade-c-hide u-hide u-js-show">
                <button type="button" class="c-facet__button c-facet__button--
border" aria-labelledby="date-legend" data-facet-expander data-facet-
target="#date-target"
                        data-track="click" data-track-action="date filter" data-
track-label="button">
                    <svg class="c-facet__icon" role="img" aria-hidden="true"
focusable="false" height="16" viewBox="0 0 16 16" width="16"
xmlns="http://www.w3.org/2000/svg"><path d="m5.58578644 3-3.29289322-
3.29289322c-.39052429-.39052429-.39052429-1.02368927 0-
1.41421356s1.02368927-.39052429 1.41421356 0l4 4c.39052429.39052429.39052429
1.02368927 0 1.41421356l-4 4c-.39052429.39052429-1.02368927.39052429-1.41421356
0s-.39052429-1.02368927 0-1.41421356z" transform="matrix(0 1 -1 0 11 3)"></path>
</svg>
                </button>
            </span>
        </legend>
        <span class="u-visually-hidden">Choose a date option to show results
from those dates only.</span>
        <div id="date-target" data-test="date-target" data-facet-
analytics="date" class="c-facet-expander u-js-hide">
            <ul class="c-facet-expander__list">

                    <li class="c-facet-expander__list-item">
                        <input name="date_range" id="date-range-today"
value="today"
                               type="radio"
                               />
                        <label class="c-facet-expander__link" for="date-range-
today">
                            <span>Today</span>
                        </label>
                    </li>

                    <li class="c-facet-expander__list-item">
                        <input name="date_range" id="date-range-last_7_days"
value="last_7_days"
                               type="radio"
                               />
                        <label class="c-facet-expander__link" for="date-range-
last_7_days">
```

```
                    <span>Last 7 days</span>
                </label>
            </li>

            <li class="c-facet-expander__list-item">
                <input name="date_range" id="date-range-last_30_days"
value="last_30_days"
                       type="radio"
                       />
                <label class="c-facet-expander__link" for="date-range-
last_30_days">
                    <span>Last 30 days</span>
                </label>
            </li>

            <li class="c-facet-expander__list-item">
                <input name="date_range" id="date-range-last_year"
value="last_year"
                       type="radio"
                       />
                <label class="c-facet-expander__link" for="date-range-
last_year">
                    <span>Last 12 months</span>
                </label>
            </li>

            <li class="c-facet-expander__list-item">
                <input name="date_range" id="date-range-last_2_years"
value="last_2_years"
                       type="radio"
                       />
                <label class="c-facet-expander__link" for="date-range-
last_2_years">
                    <span>Last 2 years</span>
                </label>
            </li>

            <li class="c-facet-expander__list-item">
                <input name="date_range" id="date-range-last_5_years"
value="last_5_years"
                       type="radio"
                       />
                <label class="c-facet-expander__link" for="date-range-
last_5_years">
                    <span>Last 5 years</span>
                </label>
            </li>

            <li class="c-facet-expander__list-item">
                <input name="date_range" id="date-range-2023-2024"
value="2023-2024"
                       type="radio"
                       checked="checked"/>
                <label class="c-facet-expander__link" for="date-range-
2023-2024">
                    <span>2023-2024</span>
                </label>
            </li>
```

```
            </ul>
            <p class="u-mb-0">
                <a class="c-facet-expander__link c-facet-expander__link--
underline" data-test="advance-search-link-date" href="/search/advanced?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-2024">Custom date
range</a>
            </p>
        </div>
    </fieldset>

                                <span class="c-facet__clear-all">
                                <a href="/search?q=llm"
                                data-track="click" data-track-category="search
results page n150"
                                data-track-action="click clear all filters"
data-track-label="(not set)">Clear all filters</a>
                            </span>
                                <div class="u-js-hide">
                                    <button type="submit" class="c-
facet__submit">
                                        <span>Apply filters</span>
                                    </button>
                                </div>
                            </div>
                            <fieldset class="c-sort-by u-pa-0" data-sort-by>
    <legend class="c-sort-by__heading">Sort by:</legend>

        <div class="c-sort-by__input-container">
            <input class="c-sort-by__input" data-sort-by-input name="order"
value="relevance" id="sort-by-relevance" type="radio" checked data-test="order-
relevance">
            <label class="c-sort-by__label" for="sort-by-
relevance">Relevance</label>
        </div>

        <div class="c-sort-by__input-container">
            <input class="c-sort-by__input" data-sort-by-input name="order"
value="date_desc" id="sort-by-date_desc" type="radio"  data-test="order-
date_desc">
            <label class="c-sort-by__label" for="sort-by-date_desc">Date
published (new to old)</label>
        </div>

        <div class="c-sort-by__input-container">
            <input class="c-sort-by__input" data-sort-by-input name="order"
value="date_asc" id="sort-by-date_asc" type="radio"  data-test="order-date_asc">
            <label class="c-sort-by__label" for="sort-by-date_asc">Date
published (old to new)</label>
        </div>

    <button class="c-sort-by__button c-facet__submit" data-sort-by-button
type="submit">Submit</button>
</fieldset>
                        </div>
```

```
<div class="c-list-header">
    <div class="u-mb-0">
        <span data-test="results-data" class="u-display-flex"><span class="u-
hide u-show-at-sm">Showing  1-50 of </span><span>668 results</span></span>
    </div>
</div>




<section id="search-article-list" class="u-mb-48 u-mt-24" data-track-
component="search grid">
    <div class="u-container">



            <ul class="app-article-list-row">

                    <li class="app-article-list-row__item">
                        <div class="u-full-height" data-native-ad-
placement="false">

    <article class="u-full-height c-card c-card--flush" itemscope
itemtype="http://schema.org/ScholarlyArticle">
        <div class="c-card__layout u-full-height">




                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs42256-024-00944-
1/MediaObjects/42256_2024_944_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs42256-024-00944-
1/MediaObjects/42256_2024_944_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
                                <img
src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs42256-024-00944-
1/MediaObjects/42256_2024_944_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>




            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s42256-024-00944-1"
                       class="c-card__link u-link-inherit"
```

```
                        itemprop="url"
                        data-track="click"
                        data-track-action="view article"

                        data-track-label="link">LLM-based agentic systems in
medicine and healthcare</a>
                </h3>


                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>Large language model-based agentic systems can
process input information, plan and decide, recall and reflect, interact and
collaborate, leverage various tools and act. This opens up a wealth of
opportunities within medicine and healthcare, ranging from clinical workflow
automation to multi-agent-aided diagnosis.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Jianing Qiu</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Kyle Lam</span></li>
<li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span
itemprop="name">Eric J. Topol</span></li>
</ul>




            </div>
        </div>


    <div class="c-card__section c-meta">
        <span class="c-meta__item c-meta__item--block-at-lg" data-
test="article.type"><span class="c-meta__type">Comments &amp; Opinion</span>
</span><time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-12-
05" itemprop="datePublished">05 Dec 2024</time>




                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">Nature Machine Intelligence</div>




                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 6, P: 1418-1420</div>
```

```
    </div>


    </article>



</div>
                    </li>

                    <li class="app-article-list-row__item">
                        <div class="u-full-height" data-native-ad-
placement="false">

    <article class="u-full-height c-card c-card--flush" itemscope
itemtype="http://schema.org/ScholarlyArticle">
        <div class="c-card__layout u-full-height">



                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs41467-024-53387-
y/MediaObjects/41467_2024_53387_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41467-024-53387-
y/MediaObjects/41467_2024_53387_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
                                <img
src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41467-024-53387-
y/MediaObjects/41467_2024_53387_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>



            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41467-024-53387-y"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">LLM-driven multimodal target
volume contouring in radiation oncology</a>
```

```
                 </h3>


                         <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                              itemprop="description">
                             <p>The integration of multimodal knowledge would be
essential for radiation oncologist to determine the therapeutic treatment. Here,
inspired by the large language models facilitating the integration of textural
information and images, this group reports a 3D multimodal clinical target
volume delineation model combining image and text-based clinical information for
decision-making in radiation oncology.</p>
                         </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Yujin Oh</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Sangjoon Park</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Jong Chul Ye</span></li>
</ul>




            </div>
        </div>


    <div class="c-card__section c-meta">
        <span class="c-meta__item c-meta__item--block-at-lg" data-
test="article.type"><span class="c-meta__type">Research</span></span><span
class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-
test="open-access"><span class="u-color-open-access">Open Access</span></span>
<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-10-24"
itemprop="datePublished">24 Oct 2024</time>




                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">Nature Communications</div>





                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 15, P: 1-14</div>



    </div>
```

```
    </article>



</div>
                    </li>

                    <li class="app-article-list-row__item">
                        <div class="u-full-height" data-native-ad-
placement="false">

    <article class="u-full-height c-card c-card--flush" itemscope
itemtype="http://schema.org/ScholarlyArticle">
        <div class="c-card__layout u-full-height">




                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs41598-024-77535-
y/MediaObjects/41598_2024_77535_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41598-024-77535-
y/MediaObjects/41598_2024_77535_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
                                <img
src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41598-024-77535-
y/MediaObjects/41598_2024_77535_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>




            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41598-024-77535-y"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Development and validation of a
novel AI framework using NLP with LLM integration for relevant clinical data
extraction through automated chart review</a>
                </h3>
```

```html
<ul data-test="author-list" class="c-author-list c-author-list--compact c-author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Mert Marcel Dagli</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Yohannes Ghenbot</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Jang W Yoon</span>
</li>
</ul>



            </div>
        </div>



    <div class="c-card__section c-meta">
        <span class="c-meta__item c-meta__item--block-at-lg" data-test="article.type"><span class="c-meta__type">Research</span></span><span class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-test="open-access"><span class="u-color-open-access">Open Access</span></span>
<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-11-05" itemprop="datePublished">05 Nov 2024</time>





                        <div class="c-meta__item c-meta__item--block-at-lg u-text-bold" data-test="journal-title-and-link">Scientific Reports</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-test="volume-and-page-info">Volume: 14, P: 1-7</div>




    </div>



    </article>




</div>
                    </li>

                    <li class="app-article-list-row__item">
                        <div class="u-full-height" data-native-ad-placement="false">
```

```
    <article class="u-full-height c-card c-card--flush" itemscope
itemtype="http://schema.org/ScholarlyArticle">
        <div class="c-card__layout u-full-height">



                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs41467-024-54457-
x/MediaObjects/41467_2024_54457_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41467-024-54457-
x/MediaObjects/41467_2024_54457_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
                                <img
src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41467-024-54457-
x/MediaObjects/41467_2024_54457_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>



            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41467-024-54457-x"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">An automatic end-to-end chemical
synthesis development platform powered by large language models</a>
                </h3>



                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>The rise of large language model (LLM) technology
offers new opportunities for advancing chemical synthesis. Here, the authors
developed an LLM-based reaction development framework (LLM-RDF) to copilot the
design and experimental tasks throughout the end-to-end chemical synthesis
development.</p>
                        </div>
```

```
<ul data-test="author-list" class="c-author-list c-author-list--compact c-
author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Yixiang Ruan</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Chenyin
Lu</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Yiming Mo</span></li>
</ul>



            </div>
        </div>



    <div class="c-card__section c-meta">
        <span class="c-meta__item c-meta__item--block-at-lg" data-
test="article.type"><span class="c-meta__type">Research</span></span><span
class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-
test="open-access"><span class="u-color-open-access">Open Access</span></span>
<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-11-23"
itemprop="datePublished">23 Nov 2024</time>





                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">Nature Communications</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 15, P: 1-16</div>




    </div>



    </article>




</div>
                    </li>

                    <li class="app-article-list-row__item">
                        <div class="u-full-height" data-native-ad-
placement="false">

    <article class="u-full-height c-card c-card--flush" itemscope
itemtype="http://schema.org/ScholarlyArticle">
        <div class="c-card__layout u-full-height">
```

```
                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs41592-024-02525-
x/MediaObjects/41592_2024_2525_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41592-024-02525-
x/MediaObjects/41592_2024_2525_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
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src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41592-024-02525-
x/MediaObjects/41592_2024_2525_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>



            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41592-024-02525-x"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Evaluation of large language
models for discovery of gene set function</a>
                </h3>



                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>Large language models show potential in
suggesting common functions for a gene set.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
author-list--truncated">
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Mengzhou Hu</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Sahar Alkhairy</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Dexter Pratt</span></li>
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            </div>
        </div>


    <div class="c-card__section c-meta">
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class="c-meta__item c-meta__item--block-at-lg" datetime="2024-11-28"
itemprop="datePublished">28 Nov 2024</time>




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text-bold" data-test="journal-title-and-link">Nature Methods</div>





                <div class="c-meta__item c-meta__item--block-at-lg" data-
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2/MediaObjects/41746_2024_1330_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41746-024-01330-
2/MediaObjects/41746_2024_1330_Fig1_HTML.png?as=webp 290w"
                                        sizes="
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                                    (max-width: 1200px) 290px,
                                    290px">
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src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41746-024-01330-
2/MediaObjects/41746_2024_1330_Fig1_HTML.png"
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            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41746-024-01330-2"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Simulating A/B testing versus
SMART designs for LLM-driven patient engagement to close preventive care
gaps</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Sanjay Basu</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Dean
Schillinger</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Joseph Rigdon</span>
</li>
</ul>




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<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-11-18"
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                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">npj Digital Medicine</div>





                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 7, P: 1-8</div>



    </div>


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</div>
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                        <div class="c-card__image">
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https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs42256-024-00925-
4/MediaObjects/42256_2024_925_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs42256-024-00925-
4/MediaObjects/42256_2024_925_Fig1_HTML.png?as=webp 290w"
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                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Contextual feature extraction
hierarchies converge in large language models and the brain</a>
                </h3>



                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>Why brain-like feature extraction emerges in
large language models (LLMs) remains elusive. Mischler, Li and colleagues
demonstrate that high-performing LLMs not only predict neural responses more
accurately than other LLMs but also align more closely with the hierarchical
language processing pathway in the brain, revealing parallels between these
models and human cognitive mechanisms.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Gavin Mischler</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Yinghao
Aaron Li</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Nima Mesgarani</span>
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    <div class="c-card__section c-meta">
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text-bold" data-test="journal-title-and-link">Nature Machine Intelligence</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 6, P: 1467-1477</div>




    </div>



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https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41591-024-03139-
8/MediaObjects/41591_2024_3139_Fig1_HTML.png?as=webp 290w"
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            <div class="c-card__body u-display-flex u-flex-direction-column">
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                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Integrated image-based deep
learning and language models for primary diabetes care</a>
                </h3>



                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>Tailored to provide diabetes management
recommendations from large training and validation datasets, an artificial
intelligence system integrating language and computer vision capabilities is
shown to improve self-management of patients in a prospective implementation
study.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Jiajia Li</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Zhouyu Guan</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Tien Yin Wong</span></li>
</ul>




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text-bold" data-test="journal-title-and-link">Nature Medicine</div>





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                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">What is in your LLM-based
framework?</a>
                </h3>
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                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
                             itemprop="description">
                            <p>To maintain high standards in clarity and
reproducibility, authors need to clearly mention and describe the use of GPT-4
and other large language models in their work.</p>
                        </div>









            </div>
        </div>



    <div class="c-card__section c-meta">
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text-bold" data-test="journal-title-and-link">Nature Machine Intelligence</div>






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    </div>



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3/MediaObjects/41598_2024_77916_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41598-024-77916-
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                       data-track-label="link">Enhancing intention prediction
and interpretability in service robots with LLM and KG</a>
                </h3>






<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Jincao Zhou</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Xuezhong Su</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Bo Liu</span></li>
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class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-
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itemprop="datePublished">06 Nov 2024</time>




                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">Scientific Reports</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-
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5/MediaObjects/41598_2024_69474_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41598-024-69474-
5/MediaObjects/41598_2024_69474_Fig1_HTML.png?as=webp 290w"
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                       data-track="click"
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                       data-track-label="link">LLM-Twin: mini-giant model-driven
beyond 5G digital twin networking framework with semantic secure communication
and computation</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact">
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itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span
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class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-
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itemprop="datePublished">17 Aug 2024</time>





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</div>
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https://media.springernature.com/w290h158/magazine-assets/d41586-024-01641-
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            <div class="c-card__body u-display-flex u-flex-direction-column">
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                       data-track-label="link">‘Fighting fire with fire’ – using
LLMs to combat LLM hallucinations</a>
                </h3>



                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
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                            <p>The number of errors produced by an LLM can be
reduced by grouping its outputs into semantically similar clusters. Remarkably,
this task can be performed by a second LLM, and the method’s efficacy can be
evaluated by a third.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact">
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                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">Nature</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-
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                       data-track-label="link">Large language models surpass
human experts in predicting neuroscience results</a>
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                        <div data-test="article-description" class="c-card__summary u-mb-16 u-hide-sm-max"
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                            <p>Large language models (LLMs) can synthesize vast amounts of information. Luo et al. show that LLMs—especially BrainGPT, an LLM the authors tuned on the neuroscience literature—outperform experts in predicting neuroscience results and could assist scientists in making future discoveries.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Xiaoliang Luo</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Akilles Rechardt</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Bradley C. Love</span></li>
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                        <div class="c-meta__item c-meta__item--block-at-lg u-text-bold" data-test="journal-title-and-link">Nature Human Behaviour</div>




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large-language model for health event prediction</a>
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<span itemprop="name">Sirui Ding</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Jiancheng Ye</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Na Zou</span></li>
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accelerate communication for eye gaze typing users with ALS</a>
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                        <div data-test="article-description" class="c-
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gaze to type and communicate. This paper presents a large language model-based
user interface that enables gaze typing in highly abbreviated forms, achieving
significant motor saving and speed gain.</p>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Shanqing Cai</span></li><li itemprop="creator"
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itemprop="name">Subhashini Venugopalan</span></li><li itemprop="creator"
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P. Brenner</span></li>
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identifying large language model outputs</a>
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                        <div data-test="article-description" class="c-
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large language models shows high text quality preservation and detection
accuracy and low latency, and is feasible in large-scale-production settings.
</p>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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See</span></li><li itemprop="creator" itemscope=""
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trials with large language models</a>
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trials. Here, the authors introduce TrialGPT, an end-to-end framework for zero-
shot patient-to-trial matching with large language models.</p>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Qiao Jin</span></li><li itemprop="creator" itemscope=""
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</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Zhiyong Lu</span></li>
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                       data-track-label="link">Evaluation of the integration of
retrieval-augmented generation in large language model for breast cancer nursing
care responses</a>
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<span itemprop="name">Ruiyu Xu</span></li><li itemprop="creator" itemscope=""
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emotional response and perception of LLM-generated questionnaire and human-
generated questionnaires</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span
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clinical oncology</a>
                </h3>






<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Nicolas Carl</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Franziska
Schramm</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Titus J.
Brinker</span></li>
</ul>




            </div>
        </div>



    <div class="c-card__section c-meta">
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itemprop="datePublished">23 Oct 2024</time>





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text-bold" data-test="journal-title-and-link">npj Precision Oncology</div>






                <div class="c-meta__item c-meta__item--block-at-lg" data-
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                    <li class="app-article-list-row__item">
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2/MediaObjects/41591_2024_3258_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41591-024-03258-
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                       data-track-label="link">A toolbox for surfacing health
equity harms and biases in large language models</a>
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                        <div data-test="article-description" class="c-card__summary u-mb-16 u-hide-sm-max"
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                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-author-list--truncated">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Stephen R. Pfohl</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Heather Cole-Lewis</span></li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Karan Singhal</span></li>
</ul>




            </div>
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                <div class="c-meta__item c-meta__item--block-at-lg" data-test="volume-and-page-info">Volume: 30, P: 3590-3600</div>




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https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41746-024-01366-
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                       data-track-label="link">Probabilistic medical predictions
of large language models</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Bowen Gu</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Rishi J. Desai</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Jie Yang</span></li>
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    <div class="c-card__section c-meta">
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itemprop="datePublished">19 Dec 2024</time>





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    </div>



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https://media.springernature.com/w290h158/springer-
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src="https://media.springernature.com/w290h158/springer-
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                    <a href="/articles/s41598-024-75599-4"
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                       data-track-label="link">Parameter-efficient fine-tuning
of large language models using semantic knowledge tuning</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Nusrat Jahan Prottasha</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Asif
Mahmud</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Ozlem Ozmen
Garibay</span></li>
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https://media.springernature.com/w290h158/springer-
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7/MediaObjects/41598_2024_60210_Fig1_HTML.png?as=webp 290w"
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                                         sizes="
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                       data-track-label="link">A multimodal approach to cross-
lingual sentiment analysis with ensemble of transformer and LLM</a>
                </h3>






<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Md Saef Ullah Miah</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Md Mohsin
Kabir</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">M. F. Mridha</span>
</li>
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LLM hallucination spreading in complex adaptive social networks</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Guozhi Hao</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Jun Wu</span></li><li
itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span
itemprop="name">Rosario Morello</span></li>
</ul>




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                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact">
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https://media.springernature.com/w290h158/springer-
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                       data-track-label="link">Reconciling the contrasting
narratives on the environmental impact of large language models</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Shaolei Ren</span></li><li itemprop="creator" itemscope=""
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</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Andrew W. Torrance</span></li>
</ul>



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    <div class="c-card__section c-meta">
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8/MediaObjects/42256_2024_832_Fig1_HTML.png?as=webp 160w,

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8/MediaObjects/42256_2024_832_Fig1_HTML.png?as=webp 290w"
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                       data-track-label="link">Augmenting large language models
with chemistry tools</a>
                </h3>



                        <div data-test="article-description" class="c-
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                            <p>Large language models can be queried to perform
chain-of-thought reasoning on text descriptions of data or computational tools,
which can enable flexible and autonomous workflows. Bran et al. developed
ChemCrow, a GPT-4-based agent that has access to computational chemistry tools
and a robotic chemistry platform, which can autonomously solve tasks for
designing or synthesizing chemicals such as drugs or materials.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Andres M. Bran</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Sam
Cox</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Philippe
Schwaller</span></li>
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            </div>
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                       data-track-label="link">Innovation and application of
Large Language Models (LLMs) in dentistry – a scoping review</a>
                </h3>






<ul data-test="author-list" class="c-author-list c-author-list--compact">
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<span itemprop="name">Fahad Umer</span></li><li itemprop="creator" itemscope=""
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                       data-track-label="link">Generative language models
exhibit social identity biases</a>
                </h3>



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                            <p>Researchers show that large language models
exhibit social identity biases similar to humans, having favoritism toward
ingroups and hostility toward outgroups. These biases persist across models,
training data and real-world human–LLM conversations.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Tiancheng Hu</span></li><li itemprop="creator"
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Kyrychenko</span></li><li itemprop="creator" itemscope=""
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                       data-track="click"
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                       data-track-label="link">Information extraction from
historical well records using a large language model</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Zhiwei Ma</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Javier E.
Santos</span></li><li itemprop="creator" itemscope=""
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                       data-track-label="link">Multimodal language and graph
learning of adsorption configuration in catalysis</a>
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                        <div data-test="article-description" class="c-
card__summary u-mb-16 u-hide-sm-max"
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                            <p>Ock and colleagues explore predictive and
generative language models for improving adsorption energy prediction in
catalysis without relying on exact atomic positions. The method involves
aligning a language model’s latent space with graph neural networks using graph-
assisted pretraining.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Janghoon Ock</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span
itemprop="name">Srivathsan Badrinarayanan</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Amir
Barati Farimani</span></li>
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                       data-track-label="link">Guidelines for ethical use and
acknowledgement of large language models in academic writing</a>
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three essential criteria for the ethical use of LLMs in academic writing, and
present a statement that researchers can quote when submitting LLM-assisted
manuscripts in order to testify to their adherence to them.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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A. Vazirani</span></li><li itemprop="creator" itemscope=""
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LLMs empowered by external tools</a>
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interest in the medical community; however, an important but overlooked aspect
of their capacity is their ability to integrate with tools. This integration
greatly extends their potential application in health care.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact">
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Nordström</span></li><li itemprop="creator" itemscope=""
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                <div class="c-meta__item c-meta__item--block-at-lg" data-
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literature using large language models</a>
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                            <p>Automated data extraction from materials science
literature using artificial intelligence and natural language processing
techniques is key to advance materials discovery. Here, the authors present a
framework to automatically extract polymer-property data from full-text journal
articles using commercially available and open-source large language models.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Sonakshi Gupta</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Akhlak
Mahmood</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Rampi Ramprasad</span>
</li>
</ul>




            </div>
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        </div>


    <div class="c-card__section c-meta">
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<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-12-19" itemprop="datePublished">19 Dec 2024</time>




                        <div class="c-meta__item c-meta__item--block-at-lg u-text-bold" data-test="journal-title-and-link">Communications Materials</div>





                <div class="c-meta__item c-meta__item--block-at-lg" data-test="volume-and-page-info">Volume: 5, P: 1-11</div>



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https://media.springernature.com/w290h158/springer-
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6/MediaObjects/43588_2024_712_Fig1_HTML.png?as=webp 290w"
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            <div class="c-card__body u-display-flex u-flex-direction-column">
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artificial intelligence</a>
                </h3>


                        <div data-test="article-description" class="c-
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                            <p>Generative artificial intelligence (GAI) is
driving a surge in e-waste due to intensive computational infrastructure needs.
This study emphasizes the necessity for proactive implementation of circular
economy practices throughout GAI value chains.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Peng Wang</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Ling-Yu Zhang</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
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https://media.springernature.com/w290h158/springer-
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program search with large language models</a>
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                            <p>FunSearch makes discoveries in established
open problems using large language models by searching for programs describing
how to solve a problem, rather than what the solution is.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Bernardino Romera-Paredes</span></li><li
itemprop="creator" itemscope="" itemtype="http://schema.org/Person"><span
itemprop="name">Mohammadamin Barekatain</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Alhussein
Fawzi</span></li>
</ul>




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text-bold" data-test="journal-title-and-link">Nature</div>





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https://media.springernature.com/w290h158/springer-
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                       data-track-label="link">Testing the reliability of an AI-
based large language model to extract ecological information from the scientific
literature</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Andrew V. Gougherty</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Hannah L.
Clipp</span></li>
</ul>




            </div>
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https://media.springernature.com/w290h158/springer-
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3/MediaObjects/41599_2024_3611_Fig1_HTML.png?as=webp 290w"
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src="https://media.springernature.com/w290h158/springer-
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                        data-track-label="link">Large language models empowered
agent-based modeling and simulation: a survey and perspectives</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Chen Gao</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Xiaochong Lan</span>
</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Yong Li</span></li>
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                       data-track-label="link">ChatMOF: an artificial
intelligence system for predicting and generating metal-organic frameworks using
large language models</a>
                </h3>



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                            <p>LLMs can be augmented with tools to increase
their capabilities. Here, authors have developed an artificial intelligence
system called ChatMOF combining LLMs and specialised libraries and utilities to
predict and generate metal-organic frameworks.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Yeonghun Kang</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Jihan
Kim</span></li>
</ul>




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                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Leveraging large language models
to improve patient education on dry eye disease</a>
                </h3>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Qais A. Dihan</span></li><li itemprop="creator"
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Brown</span></li><li itemprop="creator" itemscope=""
itemtype="http://schema.org/Person"><span itemprop="name">Abdelrahman M.
Elhusseiny</span></li>
</ul>



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    <div class="c-card__section c-meta">
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https://media.springernature.com/w290h158/springer-
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            <div class="c-card__body u-display-flex u-flex-direction-column">
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                       data-track-label="link">Bias of AI-generated content: an
examination of news produced by large language models</a>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Xiao Fang</span></li><li itemprop="creator" itemscope=""
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</li><li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Xiaohang Zhao</span></li>
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with large language models during training</a>
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                        <div data-test="article-description" class="c-
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                            <p>Prediction and interpretation tasks may be
challenging in high-stakes applications, such as medical decision-making, or
systems with compute-limited hardware. The authors introduce an augmented
framework for leveraging the knowledge learned by Large Language Models to build
interpretable models which are both accurate and efficient.</p>
                        </div>




<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Chandan Singh</span></li><li itemprop="creator"
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Askari</span></li><li itemprop="creator" itemscope=""
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language models using semantic entropy</a>
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                        <div data-test="article-description" class="c-
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model systems can be tackled by measuring uncertainty about the meanings of
generated responses rather than the text itself to improve question-
answering accuracy.</p>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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<span itemprop="name">Sebastian Farquhar</span></li><li itemprop="creator"
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reshape collective intelligence</a>
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success and is frequently supported by information technology. Burton et al.
argue that large language models are transforming information access and
transmission, presenting both opportunities and challenges for collective
intelligence.</p>
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<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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Lopez-Lopez</span></li><li itemprop="creator" itemscope=""
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                       data-track-label="link">Discovering CRISPR-Cas system
with self-processing pre-crRNA capability by foundation models</a>
                </h3>



                        <div data-test="article-description" class="c-
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                            <p>Identification of CRISPR-Cas systems by sequence
alignment is limited by homolog sequence diversity. Here, the authors develop
CHOOSER, and AI framework, for alignment free discovery of CRISPR-Cas systems,
including EphcCasλ, which was validated for self-processing and DNA cleavage.
</p>
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                       data-track-label="link">A framework for human evaluation
of large language models in healthcare derived from literature review</a>
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<span itemprop="name">Thomas Yu Chow Tam</span></li><li itemprop="creator"
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Sivarajkumar</span></li><li itemprop="creator" itemscope=""
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from scientific text with large language models</a>
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research is a complex task required specialised tools. Here the authors present
a scheme based on large language models to automatise the retrieval of
information from text in a flexible and accessible manner.</p>
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Dunn</span></li><li itemprop="creator" itemscope=""
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benefit a wide audience from different regions. Here, the authors present a
large-scale corpus, a benchmark, and a series of LLMs openly to promote
development in this field.</p>
                        </div>





<ul data-test="author-list" class="c-author-list c-author-list--compact c-
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                        <div class="c-card__image">
                            <picture>
                                <source
                                        type="image/webp"
                                        srcset="

https://media.springernature.com/w165h90/springer-
static/image/art%3A10.1038%2Fs41524-024-01449-
6/MediaObjects/41524_2024_1449_Fig1_HTML.png?as=webp 160w,

https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41524-024-01449-
6/MediaObjects/41524_2024_1449_Fig1_HTML.png?as=webp 290w"
                                        sizes="
                                    (max-width: 640px) 160px,
                                    (max-width: 1200px) 290px,
                                    290px">
                                <img
src="https://media.springernature.com/w290h158/springer-
static/image/art%3A10.1038%2Fs41524-024-01449-
6/MediaObjects/41524_2024_1449_Fig1_HTML.png"
                                     alt=""
                                     itemprop="image">
                            </picture>
                        </div>



            <div class="c-card__body u-display-flex u-flex-direction-column">
                <h3 class="c-card__title" itemprop="name headline">
                    <a href="/articles/s41524-024-01449-6"
                       class="c-card__link u-link-inherit"
                       itemprop="url"
                       data-track="click"
                       data-track-action="view article"

                       data-track-label="link">Large language models design
sequence-defined macromolecules via evolutionary optimization</a>
```

```
                </h3>




<ul data-test="author-list" class="c-author-list c-author-list--compact">
    <li itemprop="creator" itemscope="" itemtype="http://schema.org/Person">
<span itemprop="name">Wesley F. Reinhart</span></li><li itemprop="creator"
itemscope="" itemtype="http://schema.org/Person"><span itemprop="name">Antonia
Statt</span></li>
</ul>



            </div>
        </div>


    <div class="c-card__section c-meta">
        <span class="c-meta__item c-meta__item--block-at-lg" data-
test="article.type"><span class="c-meta__type">Research</span></span><span
class="c-meta__item c-meta__item--block-at-lg" itemprop="openAccess" data-
test="open-access"><span class="u-color-open-access">Open Access</span></span>
<time class="c-meta__item c-meta__item--block-at-lg" datetime="2024-11-18"
itemprop="datePublished">18 Nov 2024</time>




                        <div class="c-meta__item c-meta__item--block-at-lg u-
text-bold" data-test="journal-title-and-link">npj Computational Materials</div>





                <div class="c-meta__item c-meta__item--block-at-lg" data-
test="volume-and-page-info">Volume: 10, P: 1-8</div>




    </div>


    </article>



</div>
                    </li>

            </ul>


    </div>
</section>
```

```
            </form>
        </search>
    </div>

    <div class="u-container u-mb-48">





            <nav class="u-display-flex u-justify-content-center u-mt-24 u-mb-
24">
    <ul class="c-pagination">
        <li data-page="previous" class="c-pagination__item" data-test="page-
previous">

                <span class="c-pagination__link c-pagination__link--disabled">
                    <svg class="c-pagination__icon u-mr-8" aria-hidden="true"
height="16" viewBox="0 0 16 16" width="16"
                         xmlns="http://www.w3.org/2000/svg"><path
                            d="m4.46975946 3.28337502-4.17620792
4.03074001c-.39120768.37758093-.39160691.98937525.0000316 1.36737214.1763942
4.03091977c.39122514.3775978 1.01908149.3838182
1.40017357.0160006.39113012-.3775061.3930364-.9877245-.00310603-1.37006831-
2.48183446-
2.39538585h11.614789581.1166211-.00649339c.4973387-.055753.8833789-.46370161.883
3789-.95867408
0-.49497246-.3860402-.90292107-.8833789-.958674081-.1166211-.00649338h-
11.614789581 2.4816273-
2.39518594c.39282216-.37913917.40056173-.98637524.01946965-
1.35419292-.39113012-.37750607-1.02492687-.37784433-1.41654791.00013556z"
                            fill-rule="evenodd"/></svg><span>Previous <span
class="u-visually-hidden">page</span></span>
                </span>

        </li>




                    <li class="c-pagination__item" data-page="1" data-
test="page-1">
                    <span class="c-pagination__link c-pagination__link--active">
<span
                            class="u-visually-hidden">page </span>1</span>
                    </li>





                <li class="c-pagination__item" data-page="2" data-test="page-2">
                    <a class="c-pagination__link" href="/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-
2024&amp;page&#x3D;2">
                        <span class="u-visually-hidden">page </span>2
                    </a>
```

```
                </li>



                <li class="c-pagination__item" data-page="3" data-test="page-3">
                    <a class="c-pagination__link" href="/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-
2024&amp;page&#x3D;3">
                        <span class="u-visually-hidden">page </span>3
                    </a>
                </li>





                    <li class="c-pagination__item" data-test="page-...">
                        <span class="c-pagination__link c-pagination__link--
disabled c-pagination--ellipsis">…</span>
                    </li>





                <li class="c-pagination__item" data-page="14" data-test="page-
14">
                    <a class="c-pagination__link" href="/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-
2024&amp;page&#x3D;14">
                        <span class="u-visually-hidden">page </span>14
                    </a>
                </li>



        <li data-page="next" class="c-pagination__item" data-test="page-next">

                <a class="c-pagination__link" href="/search?
q&#x3D;llm&amp;order&#x3D;relevance&amp;date_range&#x3D;2023-
2024&amp;page&#x3D;2">
                    <span>Next <span class="u-visually-hidden">page</span>
</span>
                    <svg class="c-pagination__icon u-ml-8 c-pagination__icon--
active" aria-hidden="true"
                         height="16"
                         viewBox="0 0 16 16" width="16"
xmlns="http://www.w3.org/2000/svg">
                        <path d="m11.5302405 12.716625 4.176208-
4.03074003c.3912076-.37758093.3916069-.98937525-.0000316-1.367372l-4.1763942-
4.03091981c-.3912252-.37759778-1.0190815-.38381821-
1.4001736-.01600053-.39113013.37750607-.39303641.98772445.003106
1.3700682l2.4818345 2.39538588h-
11.6147896l-.11662112.00649339c-.49733869.055753-.88337888.46370161-.88337888.95
867408 0
.49497246.38604019.90292107.88337888.958674081.11662112.00649338h11.6147896l-
2.4816273 2.39518592c-.39282218.3791392-.40056175.9863753-.0194697
1.3541929.3911302.3775061 1.0249269.3778444 1.4165479-.0001355z"
                              fill-rule="evenodd"/>
                    </svg>
```

```
                </a>

            </li>
        </ul>
</nav>

    </div>
    <img src="/30yyyh6e/search/b3eb7dec-4f00-3eac-91de-ec64df931762" width="1"
height="1" alt="" class="visually-hidden"/>

</div>
<div id="search-menu" class="c-header__dropdown c-header__dropdown--full-width"
data-track-component="nature-150-split-header">
    <div class="c-header__container">
        <h2 class="c-header__visually-hidden">Search</h2>
        <form class="c-header__search-form" action="/search" method="get"
role="search" autocomplete="off" data-test="inline-search">
            <label class="c-header__heading" for="keywords">Search articles by
subject, keyword or author</label>
            <div class="c-header__search-layout c-header__search-layout--max-
width">
                <div>
                    <input type="text" required="" class="c-header__input"
id="keywords" name="q" value="">
                </div>
                <div class="c-header__search-layout">
                    <div>
                        <label for="results-from" class="c-header__visually-
hidden">Show results from</label>
                        <select id="results-from" name="journal" class="c-
header__select">

                                <option value="" selected>All journals</option>

                        </select>
                    </div>
                    <div>
                        <button type="submit" class="c-header__search-
button">Search</button>
                    </div>
                </div>

            </div>
        </form>

        <div class="c-header__flush">
            <a class="c-header__link" href="/search/advanced"
               data-track="click" data-track-action="advanced search" data-
track-label="link">
                Advanced search
            </a>
        </div>

        <h3 class="c-header__heading c-header__heading--keyline">Quick
links</h3>
        <ul class="c-header__list">
```

```
            <li><a class="c-header__link" href="/subjects" data-track="click"
data-track-action="explore articles by subject" data-track-label="link">Explore
articles by subject</a></li>
            <li><a class="c-header__link" href="/naturecareers" data-
track="click" data-track-action="find a job" data-track-label="link">Find a
job</a></li>
            <li><a class="c-header__link" href="/authors/index.html" data-
track="click" data-track-action="guide to authors" data-track-label="link">Guide
to authors</a></li>
            <li><a class="c-header__link" href="/authors/editorial_policies/"
data-track="click" data-track-action="editorial policies" data-track-
label="link">Editorial policies</a></li>
        </ul>
    </div>
</div>

<footer>
    <div class="u-mt-16 u-mb-16">
    <div class="u-container">
        <div class="u-display-flex u-flex-wrap u-justify-content-space-between">


            <p class="c-meta u-ma-0 u-flex-shrink">
                <span class="c-meta__item">
                    Nature.com
                </span>








            </p>
        </div>
    </div>
</div>

    <div class="c-footer">
        <div class="u-hide-print" data-track-component="footer">
    <h2 class="u-visually-hidden">nature.com sitemap</h2>
    <div class="c-footer__container">
        <div class="c-footer__grid c-footer__group--separator">
            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">About Nature Portfolio</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.nature.com/npg_/company_info/index.html"
                                                  data-track="click" data-track-
action="about us"
                                                  data-track-label="link">About
us</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.nature.com/npg_/press_room/press_releases.html"
                                                  data-track="click" data-track-
action="press releases"
```

```
                                                  data-track-label="link">Press
releases</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://press.nature.com/"
                                                  data-track="click" data-track-
action="press office"
                                                  data-track-label="link">Press
office</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://support.nature.com/support/home"
                                                  data-track="click" data-track-
action="contact us"
                                                  data-track-
label="link">Contact us</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Discover content</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/siteindex"
                                                  data-track="click" data-track-
action="journals a-z"
                                                  data-track-
label="link">Journals A-Z</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/subjects"
                                                  data-track="click" data-track-
action="article by subject"
                                                  data-track-
label="link">Articles by subject</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.protocols.io/"
                                                  data-track="click" data-track-
action="protocols.io"
                                                  data-track-
label="link">protocols.io</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.natureindex.com/"
                                                  data-track="click" data-track-
action="nature index"
                                                  data-track-label="link">Nature
Index</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Publishing policies</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.nature.com/authors/editorial_policies"
                                                  data-track="click" data-track-
action="Nature portfolio policies"
                                                  data-track-label="link">Nature
portfolio policies</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
```

```
href="https://www.nature.com/nature-research/open-access"
                                                  data-track="click" data-track-
action="open access"
                                                  data-track-label="link">Open
access</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Author &amp; Researcher
services</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/reprints"
                                                  data-track="click" data-track-
action="reprints and permissions"
                                                  data-track-
label="link">Reprints &amp; permissions</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.springernature.com/gp/authors/research-data"
                                                  data-track="click" data-track-
action="data research service"
                                                  data-track-
label="link">Research data</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://authorservices.springernature.com/language-editing/"
                                                  data-track="click" data-track-
action="language editing"
                                                  data-track-
label="link">Language editing</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://authorservices.springernature.com/scientific-editing/"
                                                  data-track="click" data-track-
action="scientific editing"
                                                  data-track-
label="link">Scientific editing</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://masterclasses.nature.com/"
                                                  data-track="click" data-track-
action="nature masterclasses"
                                                  data-track-label="link">Nature
Masterclasses</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://solutions.springernature.com/"
                                                  data-track="click" data-track-
action="research solutions"
                                                  data-track-
label="link">Research Solutions</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Libraries &amp;
institutions</h3>
```

```
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.springernature.com/gp/librarians/tools-services"
                                                  data-track="click" data-track-
action="librarian service and tools"
                                                  data-track-
label="link">Librarian service &amp; tools</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.springernature.com/gp/librarians/manage-your-
account/librarianportal"
                                                  data-track="click" data-track-
action="librarian portal"
                                                  data-track-
label="link">Librarian portal</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.nature.com/openresearch/about-open-access/information-for-
institutions"
                                                  data-track="click" data-track-
action="open research"
                                                  data-track-label="link">Open
research</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://www.springernature.com/gp/librarians/recommend-to-your-library"
                                                  data-track="click" data-track-
action="Recommend to library"
                                                  data-track-
label="link">Recommend to library</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Advertising &amp;
partnerships</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://partnerships.nature.com/product/digital-advertising/"
                                                  data-track="click" data-track-
action="advertising"
                                                  data-track-
label="link">Advertising</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://partnerships.nature.com/"
                                                  data-track="click" data-track-
action="partnerships and services"
                                                  data-track-
label="link">Partnerships &amp; Services</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"

href="https://partnerships.nature.com/media-kits/" data-track="click"
                                                  data-track-action="media kits"
data-track-label="link">Media kits</a>
                    </li>
                    <li class="c-footer__item"><a class="c-footer__link"
```

```
href="https://partnerships.nature.com/product/branded-content-native-advertising/"
                                                  data-track-action="branded
content" data-track-label="link">Branded
                        content</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Professional
development</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/naturecareers/"
                                                  data-track="click" data-track-
action="nature careers"
                                                  data-track-label="link">Nature
Careers</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://conferences.nature.com"
                                                  data-track="click" data-track-
action="nature conferences"
                                                  data-track-
label="link">Nature<span class="u-visually-hidden"> </span>
                        Conferences</a></li>
                </ul>
            </div>

            <div class="c-footer__group">
                <h3 class="c-footer__heading u-mt-0">Regional websites</h3>
                <ul class="c-footer__list">
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/natafrica"
                                                  data-track="click" data-track-
action="nature africa"
                                                  data-track-label="link">Nature
Africa</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="http://www.naturechina.com"
                                                  data-track="click" data-track-
action="nature china"
                                                  data-track-label="link">Nature
China</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/nindia"
                                                  data-track="click" data-track-
action="nature india"
                                                  data-track-label="link">Nature
India</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/natitaly"
                                                  data-track="click" data-track-
action="nature Italy"
                                                  data-track-label="link">Nature
Italy</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.natureasia.com/ja-jp"
```

```
                                                  data-track="click" data-track-
action="nature japan"
                                                  data-track-label="link">Nature
Japan</a></li>
                    <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/nmiddleeast"
                                                  data-track="click" data-track-
action="nature middle east"
                                                  data-track-label="link">Nature
Middle East</a></li>
                </ul>
            </div>

        </div>
    </div>
    <div class="c-footer__container">
        <ul class="c-footer__links">
            <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/info/privacy"
                                          data-track="click" data-track-
action="privacy policy" data-track-label="link">Privacy
                Policy</a></li>
            <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/info/cookies"
                                          data-track="click" data-track-
action="use of cookies" data-track-label="link">Use
                of cookies</a></li>
            <li class="c-footer__item">
                <button class="optanon-toggle-display c-footer__link"
onclick="javascript:;"
                        data-cc-action="preferences" data-track="click" data-
track-action="manage cookies"
                        data-track-label="link">Your privacy choices/Manage
cookies
                </button>
            </li>
            <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/info/legal-notice"
                                          data-track="click" data-track-
action="legal notice" data-track-label="link">Legal
                notice</a></li>
            <li class="c-footer__item"><a class="c-footer__link"

href="https://www.nature.com/info/accessibility-statement" data-track="click"
                                          data-track-action="accessibility
statement" data-track-label="link">Accessibility
                statement</a></li>
            <li class="c-footer__item"><a class="c-footer__link"
href="https://www.nature.com/info/terms-and-conditions"
                                          data-track="click" data-track-
action="terms and conditions"
                                          data-track-label="link">Terms &amp;
Conditions</a></li>
            <li class="c-footer__item"><a class="c-footer__link"
href="https://www.springernature.com/ccpa"
                                          data-track="click" data-track-
action="california privacy statement"
```

```
                                          data-track-label="link">Your US state
privacy rights</a></li>

        </ul>
    </div>
</div>


        <div class="c-footer__container">
    <a href="https://www.springernature.com/" class="c-footer__link">
        <img src="/static/images/logos/sn-logo-white-ea63208b81.svg"
alt="Springer Nature" loading="lazy" width="200" height="20"/>
    </a>
    <p class="c-footer__legal" data-test="copyright">&copy; 2025 Springer Nature
Limited</p>
</div>

    </div>

    <div class="u-visually-hidden" aria-hidden="true">
```

```
<?xml version="1.0" encoding="UTF-8"?><!DOCTYPE svg PUBLIC "-//W3C//DTD SVG
1.1//EN" "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd"><svg
xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink">
<defs><path id="a" d="M0 .74h56.72v55.24H0z"/></defs><symbol id="icon-access"
viewBox="0 0 18 18"><path d="m14 8c.5522847 0 1 .44771525 1 1v7h2.5c.2761424 0
.5.2238576.5.5v1.5h-18v-1.5c0-.2761424.22385763-.5.5-.5h2.5v-
7c0-.55228475.44771525-1 1-1s1 .44771525 1 1v6.9996556h8v-
6.9996556c0-.55228475.4477153-1 1-1zm-8 0 2 1v5l-2 1zm6 0v7l-2-1v-5zm-
2.42653766-7.59857636 7.03554716
4.92488299c.4162533.29137735.5174853.86502537.226108
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1.35419292-.39113012-.37750607-1.02492687-.37784433-1.41654791.00013556z" fill-
rule="evenodd"/></symbol><symbol id="icon-arrow-right" viewBox="0 0 16 16"><path
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867408 0
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1.3541929.3911302.3775061 1.0249269.3778444 1.4165479-.0001355z" fill-
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4.02412928h-2.32084805z" fill-rule="evenodd"/></symbol><symbol id="icon-block"
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transform="matrix(0 1 -1 0 11 1)"/></symbol><symbol id="icon-chevron-right"
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6z" fill-rule="evenodd"/></symbol><symbol id="icon-circle" viewBox="0 0 16 16">
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4 4 4zm0 2c-3.3137085 0-6-2.6862915-6-6s2.6862915-6 6-6 6 2.6862915 6 6-
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2.8630343 2.8630343 2.8630343c.3965875.3965876 1.0328109.3943885
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1.71428574v-6.41967746h-12v6.41967746zm10-5.3839632
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1.10185739 0-1.99508929-.8874333-1.99508929-1.999615v-6.00077002c0-
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1.10341695.90195036-1.99791312 2.0085302-1.99791312h7.9829396c1.1092806 0
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6.182zM9.812 18.222A8.41 8.41 0 109.81 1.403a8.41 8.41 0 000 16.82z" fill-
rule="evenodd"/></symbol><symbol id="icon-social-facebook" viewBox="0 0 24 24">
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2.0058587l-4.0059107-.001.001.001h-1l-.001-.001h-5l.001.001h-1l-.001-.001-
3.00391071.001c-1.10185739 0-1.99508929-.8970601-1.99508929-2.0058587v-
9.98828264c0-1.10780515.8926228-2.00585866 1.99508929-2.00585866zm-11.0059107
5h-3.999v6.9941413c0 .5572961.44630695 1.0058587.99508929
1.0058587h3.00391071zm6 0h-5v8h5zm5.0059107-4h-4.0059107v3h5.001v1h-
5.001v7.999l4.0059107.001c.5487152 0 .9950893-.4488783.9950893-1.0058587v-
9.98828264c0-.55729607-.446307-1.00585866-.9950893-1.00585866zm-12.5049107
9c.27614237 0 .5.2238576.5.5s-.22385763.5-.5.5h-1c-.27614237
0-.5-.2238576-.5-.5s.22385763-.5.5-.5zm12 0c.2761424 0
.5.2238576.5.5s-.2238576.5-.5.5h-2c-.2761424
0-.5-.2238576-.5-.5s.2238576-.5.5-.5zm-6 0c.27614237 0
.5.2238576.5.5s-.22385763.5-.5.5h-2c-.27614237
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.5.22385763.5.5s-.22385763.5-.5.5h-2c-.27614237
0-.5-.22385763-.5-.5s.22385763-.5.5-.5zm1.499-5h-5v3h5zm-6 0h-
3.00391071c-.54871518 0-.99508929.44887827-.99508929
1.00585866v1.99414134h3.999z" fill-rule="evenodd"/></symbol><symbol id="icon-
tick-circle" viewBox="0 0 24 24"><path d="m12 2c5.5228475 0 10 4.4771525 10 10s-
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1.02720949.115462 1.46437175l-6.093335
6.94397871c-.3622945.4128716-.9897871.4562317-1.4054264.0971157l-3.89719065-
3.3672071c-.42862421-.3673054-.48653564-1.0223772-.1293488-
1.4631437s.99421256-.5003185 1.42283677-.13301311l3.11097438 2.6987741z" fill-
rule="evenodd"/></symbol><symbol id="icon-tick" viewBox="0 0 16 16"><path
d="m6.76799012 9.21106946-3.1109744-2.58349728c-.42862421-.35161617-
1.06564993-.29460792-1.42283677.12733148s-.29927541 1.04903009.1293488
1.40064626l3.91576307 3.23873978c.41034319.3393961 1.01467563.2976897
1.37450571-.0948578l6.10568327-6.660841c.3613513-.41848908.3096572-
1.04610608-.115462-1.4018218-.4251192-.35571573-1.0626796-.30482786-
1.424031.11366122z" fill-rule="evenodd"/></symbol><symbol id="icon-update"
viewBox="0 0 18 18"><path d="m1 13v1c0 .5522847.44771525 1 1 1h14c.5522847 0
1-.4477153 1-1v-1h-1v-10h-14v10zm16-1h1v2c0 1.1045695-.8954305 2-2 2h-14c-
1.1045695 0-2-.8954305-2-2v-2h1v-9c0-.55228475.44771525-1 1-1h14c.5522847 0 1
.44771525 1 1zm-1 0v1h-4.5857864l-1 1h-2.82842716l-1-1h-4.58578644v-1h5l1 1h2l1-
1zm-13-8h12v7h-12zm1 1v5h10v-5zm1 1h4v1h-4zm0 2h4v1h-4z" fill-rule="evenodd"/>
</symbol><symbol id="icon-upload" viewBox="0 0 18 18"><path d="m10.0046024
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1.1092806-.8945138 2.0085302-1.9940603 2.0085302h-12.01187942c-1.10128908 0-
1.99406028-.8926228-1.99406028-1.9950893v-14.00982141c0-1.10185739.88743329-
1.99508929 1.99961498-1.99508929zm0 1h-7.00498742c-.55709576
0-.99961498.44271433-.99961498.99508929v14.00982141c0
.5500396.44491393.9950893.99406028.9950893h12.01187942c.5463747 0
.9940603-.4506622.9940603-1.0085302v-
8.98899651c0-.28393444-.2150684-.80332809-.4189472-1.0072069l-4.5763192-
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1.85576936
4.14572769c.19483374-.19483375.51177826-.19377714.70556874.0001334l2.59099082
2.59099079c.1948411.19484112.1904373.51514474.0027906.70279143-.1932998.19329987
-.5046517.19237083-.7001856-.00692852l-1.74638687-1.7800176v6.14827687c0
.2717771-.23193359.492096-.5.492096-.27614237 0-.5-.216372-.5-.492096v-
6.14827641l-1.74627892
1.77990922c-.1933927.1971171-.51252214.19455839-.70016883.0069117-.19329987-.193
29988-.19100584-.50899493.00277731-.70277808z" fill-rule="evenodd"/></symbol>
<symbol id="icon-video" viewBox="0 0 18 18"><path d="m16.0049107 2c1.1018574 0
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2.0058587-1.9950893 2.0058587h-14.00982141c-1.10185739 0-1.99508929-.8970601-
1.99508929-2.0058587v-9.98828264c0-1.10780515.8926228-2.00585866 1.99508929-
2.00585866zm0 1h-14.00982141c-.54871518 0-.99508929.44887827-.99508929
1.00585866v9.98828264c0 .5572961.44630695 1.0058587.99508929
1.0058587h14.00982141c.5487152 0 .9950893-.4488783.9950893-1.0058587v-
9.98828264c0-.55729607-.446307-1.00585866-.9950893-1.00585866zm-8.30912922
2.24944486 4.60460462 2.73982242c.9365543.55726659.9290753 1.46522435 0
2.01804082l-4.60460462 2.7398224c-.93655425.5572666-1.69578148.1645632-
1.69578148-.8937585v-5.71016863c0-1.05087579.76670616-1.446575
1.69578148-.89375851zm-.67492769.96085624v5.5750128c0
.2995102-.10753745.2442517.16578928.08477131l4.58452283-
2.67497259c.3050619-.17799716.3051624-.21655446 0-.39461026l-4.58452283-
2.67497264c-.26630747-.15538481-.16578928-.20699944-.16578928.08477139z" fill-
rule="evenodd"/></symbol><symbol id="icon-warning" viewBox="0 0 18 18"><path
d="m9 11.75c.69035594 0 1.25.5596441 1.25 1.25s-.55964406 1.25-1.25 1.25-
1.25-.5596441-1.25-1.25.55964406-1.25 1.25-1.25zm.41320045-7.75c.55228475 0
1.00000005.44771525 1.00000005 1l-.0034543.08304548-.3333333
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4c-.04586475-.55037702.36312325-1.03372649.91350028-
1.07959124l.04148683-.00259031zm-.41320045 14c-4.97056275 0-9-4.0294373-9-9 0-
4.97056275 4.02943725-9 9-9 4.9705627 0 9 4.02943725 9 9 0 4.9705627-4.0294373
9-9 9z" fill-rule="evenodd"/></symbol><symbol id="icon-checklist-banner"
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12.24 12.26c-6.76-.01-12.24-5.49-12.24-12.26zm19.33 10.66 10.23 9.22s1.21 1.09
2.3-.12l2.09-2.32s1.09-1.21-.12-2.3l-10.23-9.22m-19.29-5.92c0-4.38 3.55-7.94
7.93-7.94s7.93 3.55 7.93 7.94c0 4.38-3.55 7.94-7.93 7.94-4.38-.01-7.93-3.56-
7.93-7.94zm17.58 12.99 4.14-4.81" style="clip-
path:url(#b);fill:none;stroke:#01324b;stroke-width:2;stroke-linecap:round"/>
<path d="M8.26 9.75H28.6M8.26 15.98H28.6m-20.34 6.2h12.5m14.42-5.2V4.86s0-2.93-
2.93-2.93H4.13s-2.93 0-2.93 2.93v37.57s0 2.93 2.93 2.93h15.01M8.26
9.75H28.6M8.26 15.98H28.6m-20.34 6.2h12.5" style="clip-
path:url(#b);fill:none;stroke:#01324b;stroke-width:2;stroke-
linecap:round;stroke-linejoin:round"/></symbol><symbol id="icon-chevron-down"
viewBox="0 0 16 16"><path d="m5.58578644 3-3.29289322-
3.29289322c-.39052429-.39052429-.39052429-1.02368927 0-
1.41421356s1.02368927-.39052429 1.41421356 0l4 4c.39052429.39052429.39052429
1.02368927 0 1.41421356l-4 4c-.39052429.39052429-1.02368927.39052429-1.41421356
0s-.39052429-1.02368927 0-1.41421356z" fill-rule="evenodd" transform="matrix(0 1
-1 0 11 1)"/></symbol><symbol id="icon-eds-i-arrow-right-medium" viewBox="0 0 24
24"><path d="m12.728 3.293 7.98 7.99a.996.996 0 0 1 .281.561l.011.157c0
.32-.15.605-.384.788l-7.908 7.918a1 1 0 0 1-1.416-1.414L17.576 13H4a1 1 0 0 1 0-
2h13.598l-6.285-6.293a1 1 0 0 1-.082-1.32l.083-.095a1 1 0 0 1 1.414.001Z"/>
</symbol><symbol id="icon-eds-i-chevron-down-medium" viewBox="0 0 16 16"><path
d="m2.00087166 7h4.99912834v-4.99912834c0-.55276616.44386482-1.00087166 1-
1.00087166.55228475 0 1 .44463086 1 1.00087166v4.99912834h4.9991283c.5527662 0
1.0008717.44386482 1.0008717 1 0 .55228475-.4446309 1-1.0008717 1h-
4.9991283v4.9991283c0 .5527662-.44386482 1.0008717-1 1.0008717-.55228475 0-
1-.4446309-1-1.0008717v-4.9991283h-4.99912834c-.55276616 0-1.00087166-.44386482-
1.00087166-1 0-.55228475.44463086-1 1.00087166-1z" fill-rule="evenodd"/>
</symbol><symbol id="icon-eds-i-chevron-down-small" viewBox="0 0 16 16"><path
d="M13.692 5.278a1 1 0 0 1 .03 1.414L9.103 11.51a1.491 1.491 0 0 1-
2.188.019L2.278 6.692a1 1 0 0 1 1.444-1.384L8 9.771l4.278-4.463a1 1 0 0 1
1.318-.111l.096.081Z"/></symbol><symbol id="icon-eds-i-chevron-right-medium"
viewBox="0 0 10 10"><path d="m5.96738168 4.70639573 2.39518594-
2.41447274c.37913917-.38219212.98637524-.38972225
1.35419292-.01894278.37750606.38054586.37784436.99719163-.00013556 1.37821513l-
4.03074001 4.06319683c-.37758093.38062133-.98937525.38100976-
1.367372-.00003075l-4.03091981-
4.06337806c-.37759778-.38063832-.38381821-.99150444-.01600053-
1.3622839.37750607-.38054587.98772445-.38240057 1.37006824.00302197l2.39538588
2.4146743.96295325.98624457z" fill-rule="evenodd" transform="matrix(0 -1 1 0 0
10)"/></symbol><symbol id="icon-eds-i-chevron-right-small" viewBox="0 0 10 10">
<path d="m5.96738168 4.70639573 2.39518594-
2.41447274c.37913917-.38219212.98637524-.38972225
1.35419292-.01894278.37750606.38054586.37784436.99719163-.00013556 1.37821513l-
4.03074001 4.06319683c-.37758093.38062133-.98937525.38100976-
1.367372-.00003075l-4.03091981-
4.06337806c-.37759778-.38063832-.38381821-.99150444-.01600053-
1.3622839.37750607-.38054587.98772445-.38240057 1.37006824.00302197l2.39538588
2.4146743.96295325.98624457z" fill-rule="evenodd" transform="matrix(0 -1 1 0 0
10)"/></symbol><symbol id="icon-eds-i-chevron-up-medium" viewBox="0 0 16 16">
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1.00087166-1 0-.55228475.44463086-1 1.00087166-1z" fill-rule="evenodd"/>
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1.03281091.0006214-1.42657791-.3965122-.39651226-1.0324677-.39348875-
1.4265779.00062148l-4.2772478 4.27724782-4.27724782-
4.27724782c-.39658757-.39658757-1.03281091-.39438847-
1.42657791-.00062148-.39651226.39651227-.39348875 1.03246768.00062148
1.42657791l4.27724782 4.27724782z" fill-rule="evenodd"/></symbol><symbol
id="icon-eds-i-download-medium" viewBox="0 0 16 16"><path d="m12.9975267
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1.000316-1.0024733 1.000316h-9.99505341c-.54671233 0-1.00247329-.4478567-
1.00247329-1.000316 0-.5563109.44882258-1.000316 1.00247329-1.000316zm-
4.9975267-11.999368c.55228475 0 1 .44497754 1 .99589209v6.80214418l2.4816273-
2.48241149c.3928222-.39294628 1.0219732-.4006883
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1.41671995-.0000316l-4.17639421-4.17771394c-.39122513-.39134876-.39767006-
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1.41951349.0031070l2.48183446 2.48261871v-
6.80214418c0-.55001601.44386482-.99589209 1-.99589209z" fill-rule="evenodd"/>
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4.0294373-9-9 0-4.97056275 4.02943725-9 9-9zm0 7h-
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7 0
.47338693.32893365.86994729.77070917.97358929l.1126697.01968298.11662113.0067277
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1.25 1.25 1.25 1.25-.55964406 1.25-1.25-.55964406-1.25-1.25-1.25z" fill-
rule="evenodd"/></symbol><symbol id="icon-eds-i-mail-medium" viewBox="0 0 24
24"><path d="m19.462 0c1.413 0 2.538 1.184 2.538 2.619v12.762c0 1.435-1.125
2.619-2.538 2.619h-16.924c-1.413 0-2.538-1.184-2.538-2.619v-12.762c0-1.435
1.125-2.619 2.538-2.619zm.538 5.158-7.378 6.258a2.549 2.549 0 0 1 -3.253-.008l-
7.369-6.248v10.222c0 .353.253.619.538.619h16.924c.285 0
.538-.266.538-.619zm-.538-3.158h-16.924c-.264 0-.5.228-.534.542l8.65
7.334c.2.165.492.165.684.007l8.656-
7.342-.001-.025c-.044-.3-.274-.516-.531-.516z"/></symbol><symbol id="icon-eds-i-
menu-medium" viewBox="0 0 24 24"><path d="M21 4a1 1 0 0 1 0 2H3a1 1 0 1 1 0-
2h18Zm-4 7a1 1 0 0 1 0 2H3a1 1 0 0 1 0-2h14Zm4 7a1 1 0 0 1 0 2H3a1 1 0 0 1 0-
2h18Z"/></symbol><symbol id="icon-eds-i-search-medium" viewBox="0 0 24 24"><path
d="M11 1c5.523 0 10 4.477 10 10 0 2.4-.846 4.604-2.256 6.328l3.963 3.965a1 1 0 0
1-1.414 1.414l-3.965-3.963A9.959 9.959 0 0 1 11 21C5.477 21 1 16.523 1 11S5.477
1 11 1Zm0 2a8 8 0 1 0 0 16 8 8 0 0 0 0-16Z"/></symbol><symbol id="icon-eds-i-
user-single-medium" viewBox="0 0 24 24"><path d="M12 1a5 5 0 1 1 0 10 5 5 0 0 1
0-10Zm0 2a3 3 0 1 0 0 6 3 3 0 0 0 0-6Zm-.406 9.008a8.965 8.965 0 0 1 6.596
2.494A9.161 9.161 0 0 1 21 21.025V22a1 1 0 0 1-1 1H4a1 1 0 0 1-1-1v-.985c.05-
4.825 3.815-8.777 8.594-9.007Zm.39 1.992-.299.006c-3.63.175-6.518 3.127-6.678
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0 0 0 11.984 14Z"/></symbol><symbol id="icon-eds-i-warning-filled-medium"
viewBox="0 0 18 18"><path d="m9 11.75c.69035594 0 1.25.5596441 1.25
1.25s-.55964406 1.25-1.25 1.25-1.25-.5596441-1.25-1.25.55964406-1.25 1.25-
1.25zm.41320045-7.75c.55228475 0 1.00000005.44771525 1.00000005
1l-.0034543.08304548-.3333333
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0-.95335475-.3986624-.99654576-.91695452l-.33333333-
4c-.04586475-.55037702.36312325-1.03372649.91350028-
1.07959124l.04148683-.00259031zm-.41320045 14c-4.97056275 0-9-4.0294373-9-9 0-
4.97056275 4.02943725-9 9-9 4.9705627 0 9 4.02943725 9 9 0 4.9705627-4.0294373
9-9 9z" fill-rule="evenodd"/></symbol><symbol id="icon-expand-image" viewBox="0
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collaborate, leverage various tools and act. This opens up a wealth of
opportunities within medicine and healthcare, ranging from clinical workflow
automation to multi-agent-aided diagnosis.",
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language models (LLMs) remains elusive. Mischler, Li and colleagues demonstrate
that high-performing LLMs not only predict neural responses more accurately than
other LLMs but also align more closely with the hierarchical language processing
pathway in the brain, revealing parallels between these models and human
cognitive mechanisms.",
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of-thought reasoning on text descriptions of data or computational tools, which
can enable flexible and autonomous workflows. Bran et al. developed ChemCrow, a
GPT-4-based agent that has access to computational chemistry tools and a robotic
chemistry platform, which can autonomously solve tasks for designing or
synthesizing chemicals such as drugs or materials.",
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type and communicate. This paper presents a large language model-based user
interface that enables gaze typing in highly abbreviated forms, achieving
significant motor saving and speed gain.",
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Here, the authors introduce TrialGPT, an end-to-end framework for zero-shot
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by grouping its outputs into semantically similar clusters. Remarkably, this
task can be performed by a second LLM, and the method\u2019s efficacy can be
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of information. Luo et al. show that LLMs\u2014especially BrainGPT, an LLM the
authors tuned on the neuroscience literature\u2014outperform experts in
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for advancing nanobody development in drug discovery, but it remains a
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        "description": "Why brain-like feature extraction emerges in large language models (LLMs) remains elusive. Mischler, Li and colleagues demonstrate that high-performing LLMs not only predict neural responses more accurately than other LLMs but also align more closely with the hierarchical language processing pathway in the brain, revealing parallels between these models and human cognitive mechanisms.",
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        "title": "Stable Cox regression for survival analysis under distribution shifts",
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        "url": "/articles/s42256-024-00932-5",
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        "title": "Reshaping the discovery of self-assembling peptides with generative AI guided by hybrid deep learning",
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        "title": "An interpretable RNA foundation model for exploring functional RNA motifs in plants",
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        "url": "/articles/s42256-024-00946-z",
        "description": "Approaches are needed to explore regulatory RNA motifs in plants. An interpretable RNA foundation model is developed, trained on thousands of plant transcriptomes, which achieves superior performance in plant RNA biology tasks and enables the discovery of functional RNA sequence and structure motifs across transcriptomes.",
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        "title": "Learning spatiotemporal dynamics with a pretrained generative model",
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        "description": "Reconstructing and predicting spatiotemporal dynamics from sparse sensor data is challenging, especially with limited sensors. Li et al. address this by using self-supervised pretraining of a generative model, improving accuracy and generalization.",
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        "title": "Author Correction: A soft robot that adapts to environments through shape change",
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        "description": "Clear descriptions of intelligence in both living organisms and machines are essential to avoid confusion, sharpen thinking and guide interdisciplinary research. A Comment in this issue encourages researchers to answer key questions to improve clarity on the terms they use.",
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        "url": "/articles/s42256-024-00923-6",
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        "title": "Reusability report: Uncovering associations in biomedical bipartite networks via a bilinear attention network with domain adaptation",
        "authors": [
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        "url": "/articles/s42256-024-00822-w",
        "description": "In early 2023, Bai and colleagues presented DrugBAN, an interpretable method for drug\u2013target prediction. In this Reusability Report, Xu and colleagues reproduce the original findings and provide a careful exploration of cross-domain adaptability.",
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        "title": "Deep learning for predicting rate-induced tipping",
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risks for ecosystems and climate subsystems, yet no predictive theory existed
before. This study introduces deep learning as an effective prediction tool for
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        "title": "Multimodal language and graph learning of adsorption
configuration in catalysis",
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language models for improving adsorption energy prediction in catalysis without
relying on exact atomic positions. The method involves aligning a language
model\u2019s latent space with graph neural networks using graph-assisted
pretraining.",
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quantum computers at a computational overhead. Liao et al. demonstrate that
classical machine learning models can deliver accuracy comparable to that of
conventional techniques while reducing quantum computational costs.",
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nature. An interactive rat-like robot that engages in imitation learning with a
freely behaving rat opens a way to study social behaviours.",
        "type": "News &amp; Views"
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        "title": "Deep learning prediction of ribosome profiling with
Translatomer reveals translational regulation and interprets disease variants",
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        "title": "Fast and generalizable micromagnetic simulation with deep neural nets",
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        "url": "/articles/s42256-024-00914-7",
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        "title": "A multi-modal deep language model for contaminant removal from metagenome-assembled genomes",
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        "type": "News &amp; Views"
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        "authors": [
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        "description": "Walking efficiency declines in older adults. To address this challenge, Tricomi and colleagues present a pair of lightweight, soft robotic shorts that enhance walking efficiency for older adults by assisting leg mobility. This method improves energy efficiency on outdoor tracks while maintaining the users\u2019 natural movement control.",
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        "title": "Towards a personalized AI assistant to learn machine learning",
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        "description": "Sampling rare events is key to various fields of
science, but current methods are inefficient. Asghar and colleagues propose a
rare event sampler based on normalizing flow neural networks that requires no
prior data or collective variables, works at and out of equilibrium and keeps
efficiency constant as events become rarer.",
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\u201csubliminal techniques beyond a person\u2019s consciousness\u201d, but uses
undefined legal terms. Interdisciplinary efforts are needed to ensure effective
implementation of the legal text.",
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        "title": "Blending neural operators and relaxation methods in PDE
numerical solvers",
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        "url": "/articles/s42256-024-00910-x",
        "description": "Neural-network-based solvers for partial differential
equations (PDEs) suffer from difficulties tackling high-frequency modes when
learning complex functions, whereas for classical solvers it is more difficult
to handle low-frequency modes. Zhang and colleagues propose a hybrid numerical
PDE solver by combining a Deep Operator Network with traditional relaxation
methods, leading to balanced convergence across the eigenmode spectrum for a
wide range of PDEs.",
        "type": "Research"
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        "title": "Estimation of causal effects of genes on complex traits using
a Bayesian-network-based framework applied to GWAS data",
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but interpreting these data remains challenging. A Bayesian-network-based method
is presented that uses imputed and raw gene expression data to decipher the
causal effects of individual genes.",
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(TCR)\u2013antigen recognition remains a challenge. Zhang et al. propose a
contrastive transfer learning model to predict TCR\u2013pMHC binding that
enables interpretable analyses of epitope-specific T cells and can decipher
residue-level interactions.",
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and protein language model to generate residue sequences and full-atom
structures of protein pockets is introduced, which outperforms state-of-the-art
approaches.",
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        "title": "A plea for caution and guidance about using AI in genomics",
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        "title": "AI in biomaterials discovery: generating self-assembling
peptides with resource-efficient deep learning",
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        "url": "/articles/s42256-024-00936-1",
        "description": "Recurrent neural networks are efficient and capable
agents for discovering new peptides with strong self-organizing capabilities.",
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        "description": "The Data Provenance Initiative audits over 1,800 text artificial intelligence (AI) datasets, analysing trends, permissions of use and global representation. It exposes frequent errors on several major data hosting sites and offers tools for transparent and informed use of AI training data.",
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        "title": "A deep learning method that identifies cellular heterogeneity using nanoscale nuclear features",
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        "url": "/articles/s42256-024-00883-x",
        "description": "Cellular phenotypic heterogeneity is a key determinant of biological functions and is challenging to identify. A deep learning method that recognizes specific nuclear signatures is discussed, which can identify cellular heterogeneity and differentiate between various cell states using a small amount of super-resolution microscopy data.",
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        "url": "/articles/s41467-024-54368-x",
        "description": "It remains unclear whether rRNA modifications can be naturally altered in response to antibiotics in bacteria. Here, the authors analyzed direct RNA nanopore sequencing data with an analytical pipeline <i>Nanoconsensus</i>, to investigate whether bacterial rRNA modifications are modulated upon exposure to various antibiotics.",
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        "title": "Natural variation in the promoter of <i>qRBG1</i>/<i>OsBZR5</i> underlies enhanced rice yield",
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        "authors": [
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        "title": "Natural variation of <i>TBR</i> confers plant zinc toxicity tolerance through root cell wall pectin methylesterification",
        "authors": [
          "Kaizhen Zhong",
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        "url": "/articles/s41467-024-50106-5",
        "description": "Zinc is an essential micronutrient for plants, however its excess causes cellular damage. Here, the authors report that natural variation of <i>Trichome Birefringence</i> (<i>TBR</i>) gene confers zinc toxicity tolerance through modulating root cell wall pectin methylesterification in Arabidopsis.",
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        "title": "Natural variation in <i>GmSW17</i> controls seed size in soybean",
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        "url": "/articles/s41467-024-51798-5",
        "description": "Seed size plays an important role in determining soybean yield. Here, the authors report <i>Gm</i><i>SW17</i>, encoding a homolog of <i>Arabidopsis</i> UBP22 that plays a role in deubiquitination, as a positive regulator of soybean seed width and seed weight through inhibition of the G1-to-S transition by interacting with GmSGF11 and GmENY2.",
        "type": "Research"
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        "title": "Native mass spectrometry and structural studies reveal modulation of MsbA\u2013nucleotide interactions by lipids",
        "authors": [
          "Tianqi Zhang",
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        "url": "/articles/s41467-024-50350-9",
        "description": "MsbA mediates the transport of lipopolysaccharide across the inner membrane. Here, the authors show specific MsbA-lipid interactions can tune the selectivity for binding ATP over ADP and present five structures of MsbA with one bound to lipid.",
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        "title": "Natural variation in <i>BnaA07.MKK9</i> confers resistance to Sclerotinia stem rot in oilseed rape",
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        "url": "/articles/s41467-024-49504-6",
        "description": "The genetic basis underlying resistance to Sclerotinia stem rot (SSR) in oilseed rape remains elusive. Here, the authors identify BnaA07.MKK9 as a pivotal regulator of SSR resistance in oilseed rape by GWAS, providing new insights into plant defense mechanisms against necrotrophic pathogens.",
        "type": "Research"
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        "title": "Natural reversion promotes LPS elongation in an attenuated <i>Coxiella burnetii</i> strain",
        "authors": [
          "Carrie M. Long",
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        ],
        "url": "/articles/s41467-023-43972-y",
        "description": "In vitro propagation of the pathogenic bacterium <i>Coxiella burnetii</i>, the causative agent of Q fever, leads to attenuated virulence and lipopolysaccharide (LPS) truncation. Here, Long et al. show that a strain considered to be avirulent (NMII) can be recovered from infected animals, and these isolates display increased virulence and an elongated LPS due to reversion of a 3-bp mutation in a gene.",
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        "title": "Native N-glycome profiling of single cells and ng-level blood isolates using label-free capillary electrophoresis-mass spectrometry",
        "authors": [
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        "url": "/articles/s41467-024-47772-w",
        "description": "Single-cell manipulation and processing techniques and improvements in mass spectrometry sensitivity make single-cell proteomic profiling feasible. This study presents a label-free approach for the characterisation of native N-glycans of single mammalian cells and ng-level blood isolates, demonstrating the potential to detect cell surface glycome changes at the single-cell level in health or disease.",
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        "title": "Natural variation in <i>OsMYB8</i> confers diurnal floret opening time divergence between <i>indica</i> and <i>japonica</i> subspecies",
        "authors": [
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        "description": "Florets of <i>indica</i> rice open earlier than <i>japonica</i> rice, hindering utilization of the cross subspecies heterosis. Here, the authors show that an <i>OsMYB8-OsJAR1</i> module regulates diurnal floret opening time divergences between the two subspecies.",
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        "title": "Native-state proteomics of Parvalbumin interneurons identifies unique molecular signatures and vulnerabilities to early Alzheimer\u2019s pathology",
        "authors": [
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        "url": "/articles/s41467-024-47028-7",
        "description": "Native state proteomics of PV interneurons revealed unique molecular features of high translational and metabolic activity, and enrichment of Alzheimer\u2019s risk genes. Early amyloid pathology exerted unique effects on mitochondria, mTOR signaling and neurotransmission in PV neurons.",
        "type": "Research"
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        "title": "Natural variation of STKc_GSK3 kinase TaSG-D1 contributes to heat stress tolerance in Indian dwarf wheat",
        "authors": [
          "Jie Cao",
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        ],
        "url": "/articles/s41467-024-46419-0",
        "description": "Indian dwarf wheat (<i>Triticum sphaerococcum</i>) is thermotolerant, but the underlying mechanism is unclear. Here, the authors report the cloning of the heat tolerance gene encoding a STKc_GSK3 kinase and its variation affects phosphorylation level of downstream TaPIF4 in determining thermotolerance.",
        "type": "Research"
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        "title": "Natural selection and genetic diversity maintenance in a parasitic wasp during continuous biological control application",
        "authors": [
          "Bingyan Li",
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        "url": "/articles/s41467-024-45631-2",
        "description": "Parasitoid wasps are reared and released as biocontrol agents to manage aphids and protect crops. Here, the authors use genomes from 542 wasps to show that, in spite of wide scale release of low-diversity captive individuals, diversity in wild populations is maintained.",
        "type": "Research"
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        "title": "Natural diversity screening, assay development, and characterization of nylon-6 enzymatic depolymerization",
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        "description": "Polyamides (PAs) or nylons are types of plastics with wide applications, but due to their accumulation in the environment, strategies for their deconstruction are of interest. Here, the authors screen 40 potential nylon-hydrolyzing enzymes (nylonases) using a mass spectrometry-based approach and identify a thermostabilized N-terminal nucleophile hydrolase as the most promising for further development, as well as crucial targets for progressing PA6 enzymatic depolymerization.",
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        "url": "/articles/s41467-024-46271-2",
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        "description": "On the 12th of April 2010, <i>Nature Communications</i> published its first <a href=\"https://www.nature.com/articles/ncomms1011\">editorial&lt;/a&gt; and primary research articles. The topics of these first 11 papers represented the multidisciplinary nature of the journal: from <a href=\"https://www.nature.com/articles/ncomms1003\">DNA damage&lt;/a&gt; to <a href=\"https://www.nature.com/articles/ncomms1010\">optics&lt;/a&gt; alongside <a href=\"https://www.nature.com/articles/ncomms1006\">material science&lt;/a&gt; to <a href=\"https://www.nature.com/articles/ncomms1000\">energy&lt;/a&gt; and including <a href=\"https://www.nature.com/articles/ncomms1004\">polymer chemistry&lt;/a&gt;. We have spoken with the corresponding authors of some of these very first papers and asked them about their experience of publishing in this then new journal and how they see <i>Nature Communications</i> now.",
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        "title": "Small-angle scattering interferometry with neutron orbital angular momentum states",
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orbital angular momentum (OAM) in small-angle neutron scattering experiments
provides novel approaches to the characterisation of material properties. Here
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        "title": "Aza-[4\u2009+\u20092]-cycloaddition of benzocyclobutenones
into isoquinolinone derivatives enabled by photoinduced regio-specific C\u2013C
bond cleavage",
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mild reaction conditions remains a challenge. Herein, the authors report a
photoinduced catalyst-free regio-specific bond cleavage of benzocyclobutenones,
enabling the generation of ortho-quinoid ketene methides for aza-
[4\u2009+\u20092]-cycloaddition with imines. Catalytic asymmetric version is
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is emerging as a valuable tool for the development of light-based technologies.
Here the authors harness time-dependent non-Hermitian Hamiltonians to
demonstrate dynamic control over the skin effect.",
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        "title": "Infinitesimal optical singularity ruler for three-dimensional
picometric metrology",
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This ruler is capable of measuring three-dimensional picometric displacements
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        "url": "/articles/s41467-024-55126-9",
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        "url": "/articles/s41467-024-55117-w",
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        "title": "Palladium-catalyzed remote internal C(<i>sp</i><sup>3</sup>)\u2212H bond chlorination of alkenes",
        "authors": [
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"description": "C(sp³)\u2013Cl bonds are present in numerous biologically active molecules and can also be used as a site for diversification by substitution or cross-coupling reactions. Here, the authors report a remote internal site-selective C(sp³)\u2013H bond chlorination of alkenes through sequential alkene isomerization and hydrochlorination, enabling the synthesis of both benzylic and tertiary chlorides with excellent site-selectivity.",
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"description": "Hydrogels with room temperature phosphorescence have potential in a number of applications, but mechanical properties can limit the potential. Here, the authors report a wood-based hydrogel with room temperature phosphorescence, by polymerization of acrylamide with delignified wood.",
"type": "Research"
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{
"title": "Global lake phytoplankton proliferation intensifies climate warming",
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"Wenqing Shi",
"Boqiang Qin",
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"url": "/articles/s41467-024-54926-3",
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"type": "Research"
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"title": "N-acetyltransferase 10 is implicated in the pathogenesis of cycling T cell-mediated autoimmune and inflammatory disorders in mice",
"authors": [
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"url": "/articles/s41467-024-53350-x",
"description": "Abnormal T cell proliferation often triggers autoimmune disorders. Here the authors show that T cell-specific deficiency of the enzyme N-acetyltransferase 10 ameliorates experimental autoimmune encephalitis potentially by RACK1-mediated regulation of T cell metabolism and expansion.",
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"title": "Creating glycoside diversity through stereoselective carboboration of glycals",
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diversifiable positions from readily available sugar precursors remains a
challenging task. Herein, the authors present a regio- and stereoselective
nickel-catalyzed carboboration of glycals, which offers a platform for
generating glycoside diversity with diverse C1 and C2 modification potential.",
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layers of materials of various surfaces when making e.g., electronic devices.
Here authors show how addition of a heavy inert gas can increase the thickness
uniformity of the films on complex objects.",
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disease onset and enables the rapid response to emerging health threats",
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capability for small molecule activation but their application in redox
catalysis remains elusive. Here, the authors report the synthesis and isolation
of a stannylene with carbodiphosphorane ligand and characterize its catalytic
activity in the hydrodefluorination reaction of fluoroarenes.",
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but traditionally require external electron donors for activation. Herein, the
authors report an energy transfer strategy for the activation of aryl
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thin-film lithium niobate waveguides",
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infimum in the topological pumping process using lithium niobate-on-insulator
waveguides, exceeding the limitations of conventional pumps. These results will
facilitate the design of compact topological devices.",
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        "title": "Enantioselective construction of C-B axially chiral
alkenylborons by nickel-catalyzed radical relayed reductive coupling",
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alkenes remains a challenge due to the lower rotational barrier, especially for
longer stereogenic axis. Here the authors report a nickel-catalyzed
atroposelective radical relayed reductive coupling reaction of ethynyl-
azaborines with alkyl/aryl halides through a boron-stabilized vinyl radical
intermediate.",
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        "url": "/articles/s41467-024-54851-5",
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skyrmions in [Pt/Co]<sub>3</sub>/Ru/[Co/Pt]<sub>3</sub> multilayers using spin-
orbit torque and thermal effects, enabling motion reversal. This method advances
skyrmion-based spintronic devices for data storage and quantum computing.",
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        "url": "/articles/s41467-024-51005-5",
        "description": "C-terminal \u03b1-amidated peptides are attractive therapeutic targets, but preparative methods to access amidated pharmaceuticals are limited both on lab and manufacturing-scale. Here, the authors report a straightforward and scalable approach to the C-terminal \u03b1-amidation of peptides and proteins from cysteine-extended polypeptide precursors.",
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        "description": "The photochemical transformations of aryl thiols to
other functional groups have been scarcely explored. Here the authors present a
carboxylation of aryl thiols using 1 atmosphere of CO<sub>2</sub> under
photoirradiative conditions, a methodology which can be extended to the
degradation of polyphenylene sulfide.",
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nanospace of a densely fluorinated metal-organic framework",
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integration of adsorption and dissolution, so-called \u201cadsorptive-
dissolution\u201d, which enables selective adsorption of O<sub>2</sub> over
Ar.",
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          "Yuanli Xu",
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        "url": "/articles/s41467-024-53728-x",
        "description": "Incorporation of deuterium into small molecules in the
place of hydrogens can alter the properties of the molecule in ways that provide
benefits without affecting the core function of the molecule. Here, the authors
report a ring-opening deuteration of cyclic alkanes via biphosphonium
photocatalysis.",
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        "title": "Growing large single crystals of two- or three-dimensional
covalent organic polymers through unconventional Te-O-P linkages",
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          "Miaomiao Xue",
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        "description": "Understanding precise structures of two-/three-
dimensional covalent organic polymers through single-crystal X-ray diffraction
analysis is interesting however, to grow high-quality single crystals of these
materials is challenging. Here, the authors incorporate tellurium into the
backbones covalent organic polymers as a concise and fast method to grow large
single crystals.",
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        "title": "High sensitivity cameras can lower spatial resolution in high-resolution optical microscopy",
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        "title": "Direct oxygen insertion into C-C bond of styrenes with air",
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        "description": "The direct oxygenation of C\u2013C bonds through single-oxygen-atom insertion like the Baeyer-Villiger reaction is of particular significance. Herein, the authors present an approach for the skeletal modification of styrene using O<sub>2</sub> via oxygen insertion, resulting in the formation of aryl ether frameworks under mild reaction conditions.",
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        "title": "Visible light-mediated organocatalyzed 1,3-aminoacylation of cyclopropane employing N-benzoyl saccharin as bifunctional reagent",
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        "url": "/articles/s41467-024-53202-8",
        "description": "The carboamination of unsaturated molecules with bifunctional carboamination reagents is considered to be an attractive approach for the synthesis of nitrogen-containing compounds but employing bifunctional C\u2013N reagents in the functionalization of cyclopropane remains underdeveloped. Here the authors present N-heterocyclic carbenes and a photoredox catalyst dualcatalyzed 1,3-aminoacylation of cyclopropane employing N-benzoyl saccharin as a bifunctional reagent.",
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        "title": "High drain field impact ionization transistors as ideal
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        "description": "Yuan et al. report a nearly vertical subthreshold swing
field-effect transistor consists of a graphene/silicon heterojunction drain and
a silicon channel. The device enables nearly hysteresis-free transistors with
subthreshold swing of 16 \u00b5V dec<sup>\u22121</sup>, and a complementary
logic inverter with gain of 311.",
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        "title": "Role of anthropogenic forcing in Antarctic sea ice variability
simulated in climate models",
        "authors": [
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        "url": "/articles/s41467-024-54485-7",
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Southern Hemisphere atmospheric variability, the Southern Annular Mode, plays a
more significant role in Antarctic sea ice variability than the deep convection
in the Southern Ocean.",
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        "title": "Main-group compounds selectively activate natural gas alkanes
under room temperature and atmospheric pressure",
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        "description": "Activation of natural gas alkanes using main-group
systems typically requires heating or photo-irradiation under high atmospheric
pressure with low regioselectivity. Here the authors report Lewis acid-carbene
adduct activation of C\u2013H bonds in ethane, propane and <i>n</i>-butane under
one atmospheric pressure and at room temperature with high selectivity.",
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electron-deficient olefins and its application to the synthesis of lucidumone",
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between two electron-deficient components is synthetically useful and yet
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        "title": "Robust bendable thermoelectric generators enabled by
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        "description": "The authors strengthen the elasticity of
Ag<sub>2</sub>Se by microstructure engineering approach, realizing bendability
of the thermoelectric generator for at least 1,000,000 times at a tiny bending
radius of 3\u2009mm with a high power density.",
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        "title": "NAT10-dependent <i>N</i><sup>4</sup>\u2010acetylcytidine
modification mediates PAN RNA stability, KSHV reactivation, and IFI16-related
inflammasome activation",
        "authors": [
          "Qin Yan",
          "Jing Zhou",
          "Chun Lu"
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        "url": "/articles/s41467-023-42135-3",
        "description": "<i>N</i>-acetyltransferase 10 (NAT10) is an <i>N</i>
<sup>4</sup>\u2010acetylcytidine (ac<sup>4</sup>C) writer, which catalyzes RNA
acetylation at cytidine N<sup>4</sup> position on RNAs. Here, the authors show
that NAT10 catalyzes ac<sup>4</sup>C addition to a long non-coding RNA encoded
by the oncogenic DNA virus Kaposi\u2019s sarcoma-associated herpesvirus (KSHV),
triggering viral lytic reactivation from latency, which promotes NAT10
recruitment of IFI16 mRNA, resulting in inflammasome activation.",
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        "title": "National quantifications of methane emissions from fuel
exploitation using high resolution inversions of satellite observations",
        "authors": [
          "Lu Shen",
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          "Jintai Lin"
        ],
        "url": "/articles/s41467-023-40671-6",
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"description": "High-resolution satellite data enables a unique verification of national methane emissions worldwide. Global estimates are 63 Tg a^{\u22121} for oil-gas, 30% higher than the UNFCCC reports due to under-reporting by four largest emitters, and 33 Tg a^{\u22121} for coal, consistent with previous estimates.",
"type": "Research"
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"title": "Natural resources modulate the nexus between environmental shocks and human mobility",
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"url": "/articles/s41467-023-37074-y",
"description": "Remote sensed information and population data for continental Africa are used to assess how migration acts as an adaptation response after drought event. The effect on mobility is amplified with drought frequency and poverty.",
"type": "Research"
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"title": "National surveillance data analysis of COVID-19 vaccine uptake in England by women of reproductive age",
"authors": [
"Laura A. Magee",
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"url": "/articles/s41467-023-36125-8",
"description": "Women of reproductive age may have specific concerns relating to perceived impacts on fertility and menstrual cycles that make them hesitant to receive COVID-19 vaccination. In this study, the authors explore COVID-19 vaccine uptake rates in women of reproductive age using linked data for ~13 million women in England.",
"type": "Research"
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{
"title": "Native qudit entanglement in a trapped ion quantum processor",
"authors": [
"Pavel Hrmo",
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"url": "/articles/s41467-023-37375-2",
"description": "Encoding quantum information in qudits instead of qubits allows for several advantages, but scalable native entangling techniques would be needed. Here, the authors show how to use light-shift gates to perform entangling operations on trapped ion systems, with a calibration overhead which is independent on the qudit dimension.",
"type": "Research"
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{
"title": "National genomic surveillance integrating standardized quantitative susceptibility testing clarifies antimicrobial resistance in Enterobacterales",
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          "Shizuo Kayama",
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        "description": "Kayama <i>et al</i>. present a blueprint for a national genomic surveillance study that conducts genome sequencing of thousands of strains, integrates standardized quantitative antimicrobial susceptibility testing, and characterizes antimicrobial resistance determinants.",
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        "title": "Natural oxidase-mimicking copper-organic frameworks for targeted identification of ascorbate in sensitive sweat sensing",
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        "url": "/articles/s41467-022-35721-4",
        "description": "Sweat sensors are important in personalized healthcare using natural oxidase to target biomolecules but these reactions are susceptible to external interference. Here, the authors report tryptophan- and histidine-treated copper metal-organic frameworks which show highly selective activity for ascorbate oxidation and can serve as an efficient ascorbate oxidase-mimicking material in sensitive sweat sensors.",
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        "title": "Natural product P57 induces hypothermia through targeting pyridoxal kinase",
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        "url": "/articles/s41467-023-41435-y",
        "description": "Induction of hypothermia during hibernation/torpor enables certain mammals to survive under extreme conditions. Here, the authors show that the natural product P57 induces hypothermia by targeting pyridoxal kinase and has a potential application in therapeutic hypothermia.",
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        "title": "Natural plant growth and development achieved in the IPK PhenoSphere by dynamic environment simulation",
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        "url": "/articles/s41467-023-41332-4",
        "description": "The PhenoSphere is a unique plant cultivation facility in which field-like environments can be simulated. Here, the authors find that a single season simulation is superior to an averaged season and to a climatized glasshouse cultivation to elicit field-like phenotypes evaluated in 11 maize lines.",
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        "title": "Natural statistics support a rational account of confidence biases",
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        "url": "/articles/s41467-023-39737-2",
        "description": "Human decision confidence displays a number of biases and has been shown to dissociate from decision accuracy. Here, by using neural network and Bayesian models, the authors show that these effects can be explained by the statistics of sensory inputs.",
        "type": "Research"
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        "title": "Native American ataxia medicines rescue ataxia-linked mutant potassium channel activity via binding to the voltage sensing domain",
        "authors": [
          "R\u00Oedan W. Manville",
          "J. Alfredo Freites",
          "Geoffrey W. Abbott"
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        "url": "/articles/s41467-023-38834-6",
        "description": "Drugs that rescue function of episodic ataxia 1 (EA1) mutant potassium channels are lacking. Here, Manville et al identify and describe the molecular basis for Native American botanical ataxia remedies that directly rescue EA1 mutant channels.",
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        "title": "Native doublet microtubules from <i>Tetrahymena thermophila</i> reveal the importance of outer junction proteins",
        "authors": [
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        "url": "/articles/s41467-023-37868-0",
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        "description": "Using cryo-EM, the authors identified 42 MIPs, including outer junction protein CFAP77 and outer dense fibers, in native doublet microtubules of <i>Tetrahymena thermophila</i>. Knockout of CFAP77 reduced ciliary beat frequency and led to outer junction damage.",
        "type": "Research"
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        "title": "Natural killer cells and innate lymphoid cells 1 tune anxiety-like behavior and memory in mice via interferon-\u03b3 and acetylcholine",
        "authors": [
          "Stefano Garofalo",
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        "url": "/articles/s41467-023-38899-3",
        "description": "The meningeal compartment communicates with the brain to modulate homeostatic functions. Here, the authors demonstrate that natural killer (NK) cells and innate lymphoid cells (ILC) 1 shape synaptic neuronal transmission and affect mouse behavior.",
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        "title": "Native structure of mosquito salivary protein uncovers domains relevant to pathogen transmission",
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        "url": "/articles/s41467-023-36577-y",
        "description": "Female mosquitoes inject saliva into vertebrate hosts during blood feeding, transmitting mosquito-borne pathogens. Here, cryo-EM of mosquito salivary gland extract uncovers the native SGS1 structure and domains relevant to pathogen transmission.",
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        "title": "Maternal NAT10 orchestrates oocyte meiotic cell-cycle progression and maturation in mice",
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        "url": "/articles/s41467-023-39256-0",
        "description": "Generation of mature oocytes requires tight regulation of a discontinuous meiotic cell cycle. Here they show that the acetyltransferase Nat10 mediates modification of RNAs targeted for degradation and find that this process is essential for female oocyte meiosis and maturation.",
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        "url": "/articles/s41467-023-41148-2",
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        "title": "The oxygen-tolerant reductive glycine pathway assimilates methanol, formate and CO<sub>2</sub> in the yeast <i>Komagataella phaffii</i>",
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        "url": "/articles/s41467-023-43610-7",
        "description": "One carbon compounds such as CO<sub>2</sub>, methanol and formate are cost-effective and environmentally friendly microbial feedstocks for biomanufacturing. Here, the authors report the oxygen tolerant reductive glycine pathway in <i>Komagataella phaffii</i> can co-assimilate CO<sub>2</sub>, methanol and formate.",
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        "title": "Embedding security into ferroelectric FET array via in situ memory operation",
        "authors": [
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        "url": "/articles/s41467-023-43941-5",
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        "title": "Direct probing of single-molecule chemiluminescent reaction dynamics under catalytic conditions in solution",
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        "title": "3D printing of self-healing personalized liver models for surgical training and preoperative planning",
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        "url": "/articles/s41467-023-44324-6",
        "description": "High-fidelity personalized anatomical models can offer invaluable support for precision medicine. Here, the authors show the 3D printing of self-healing liver models, which can be resected in a trial and-error manner for surgical training to enhance the safety of hepatic surgery.",
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twisted graphene multilayers, but these structures are typically not
energetically favorable. This study extends these observations to Bernal-stacked
tetralayer graphene, which is the most stable configuration of four-layer
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        "title": "Phonon-enhanced nonlinearities in hexagonal boron nitride",
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        "url": "/articles/s41467-023-43501-x",
        "description": "Nonlinear optical processes like higher-order harmonic
generation in solids depend on several factors. Here the authors explore the
optical nonlinearity of hexagonal boron nitride and find that enhanced
nonlinearity is due to electron-phonon and phonon-polariton couplings.",
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interface is key to understanding the catalytic reaction mechanism. Here, the
authors develop a STM-BJ method to monitor the catalytic process from the
perspective of single-bond energy.",
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        "title": "Evidence for two dimensional anisotropic Luttinger liquids at
millikelvin temperatures",
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        "description": "Recently, a Luttinger liquid state was reported in a moir\u00e9 superlattice of bilayer tungsten ditelluride at small twist angles and temperatures of a few kelvins. Here, the authors extend this result to millikelvin temperatures, supporting the existence of the 2D anisotropic Luttinger liquid as a stable ground state.",
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        "title": "Migration direction in a songbird explained by two loci",
        "authors": [
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        "url": "/articles/s41467-023-35788-7",
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        "title": "Structure of native chromatin fibres revealed by Cryo-ET in situ",
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          "Frank Nightingale",
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        "url": "/articles/s41467-023-42072-1",
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        "title": "A recyclable polyester library from reversible alternating copolymerization of aldehyde and cyclic anhydride",
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        "title": "Decarboxylation of <i>\u03b2</i>-boryl NHPI esters enables radical 1,2-boron shift for the assembly of versatile organoborons",
        "authors": [
          "Yu Guo",
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        "description": "The 1,2 migration reaction on tetracoordinate boron species have been well investigated but less work looked on 1,2-boryl radical shift for the construction of organoborons. Here, the authors demonstrate a visible-light-induced photoredox neutral decarboxylative radical cross-coupling reaction, which undergoes a radical 1,2-boron shift to give a translocated C-radical for further capture of versatile radical acceptors",
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        "url": "/articles/s41467-023-40801-0",
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        "title": "Catalytic asymmetric dearomatization of phenols via divergent intermolecular (3\u2009+\u20092) and alkylation reactions",
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          "Xiang Gao",
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        "url": "/articles/s41467-023-40891-w",
        "description": "The catalytic asymmetric dearomatization (CADA) reaction is a powerful protocol for the assembly of three-dimensional cyclic compounds but phenols have been considered challenging substrates. Here, the authors report the chiral phosphoric acidcatalyzed divergent intermolecular CADA reactions of phenols with azoalkenes to obtain tetrahydroindolone and cyclohexadienone products in good yields with excellent <i>ee</i> values.",
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        "title": "Identifying <i>s</i>-wave pairing symmetry in single-layer FeSe from topologically trivial edge states",
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        "url": "/articles/s41467-023-40931-5",
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        "title": "Cascaded metasurfaces for high-purity vortex generation",
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        "description": "The authors demonstrate an efficient way to generate
high-purity vortex beams by applying optical neural networks to cascaded phase-
only metasurfaces. Specifically, they present record-high-quality Laguerre-
Gaussian (<i>LG</i><sub><i>p,l</i></sub>) optical modes with polynomial orders
<i>p</i>\u2009=\u200910 and <i>l</i>\u2009=\u2009200 with purity in <i>p</i>,
<i>l</i> and relative conversion efficiency of 96%, 85%, and 70%,
respectively.",
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European hops calls for immediate adaptation measures",
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decline in the quality and quantity of traditional aroma hops across Europe and
calls for urgent adaptation measures to stabilize international market chains.",
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        "title": "Interlayer donor-acceptor pair excitons in
MoSe<sub>2</sub>/WSe<sub>2</sub> moir\u00e9 heterobilayer",
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moir\u00e9 superlattices is still elusive Here, the authors propose a donor-
acceptor pair mechanism for LIXs in MoSe<sub>2</sub>/WSe<sub>2</sub>
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polarization-independent, broadband edge detection with high transmission
efficiency under both coherent and incoherent illumination along the visible
range using a metasurface based on Fourier optics principles.",
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        "title": "Fast viral dynamics revealed by microsecond time-resolved cryo-EM",
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        "url": "/articles/s41467-023-41444-x",
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        "title": "Global tree growth resilience to cold extremes following the Tambora volcanic eruption",
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        "url": "/articles/s41467-023-42409-w",
        "description": "Explosive volcanic eruptions cause abrupt global cooling as happened after the 1809 and 1815 Tambora eruptions. Here, the authors show how forest growth was severely impacted by such cold extremes in high latitudes and elevations and that recovery took longer in mid-latitude regions.",
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        "title": "All-silicon quantum light source by embedding an atomic
emissive center in a nanophotonic cavity",
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        "url": "/articles/s41467-023-38559-6",
        "description": "The use of silicon for integrated quantum photonic
technologies is currently hindered by the lack of suitable on-demand quantum
light sources. Here, the authors fill this gap by demonstrating the creation of
single atomic emissive centers in silicon and their efficient coupling with
nanophotonic cavities.",
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can be modulated at 75 MHz. Moreover, the repetition rate can be locked to an
external microwave reference by direct injection locking or feedback locking
without external modulation.",
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        "title": "A 13-million turnover-number anionic Ir-catalyst for a
selective industrial route to chiral nicotine",
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        "url": "/articles/s41467-023-39375-8",
        "description": "The development of catalysts for practical asymmetric
hydrogenation of ketones remains an important goal of synthetic organic
chemistry. Here, an anionic iridium catalyst with excellent activity is reported
and used in a hundred-kilogram-scale reduction as part of a route to chiral
nicotine.",
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thin films",
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        "url": "/articles/s41467-023-40757-1",
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microscopy and muon spin rotation. They find an anomalous absence of Meissner
screening near the Pt/Nb interface due to spin-triplet pair correlations driven
by spin-orbit coupling alone with no ferromagnetic layer necessary.",
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        "title": "A look back at embryo models",
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development as our Method of the Year. Here, we catch up on what has happened in
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over a wide range of imaging conditions including light and electron
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chromatin accessibility, histone modifications and targeted proteins.",
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practice and find it empowers science and collaboration.",
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proteins per single cell at unseen accuracy to uncover cellular heterogeneity",
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the Orbitrap Astral mass spectrometer.",
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spatial omics modalities, despite differences in data quality and spatial
resolution for improved feature extraction and clustering to reveal biologically
meaningful tissue organization.",
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to five omic modalities in situ.",
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control of cellular functions deep within tissues, although few such proteins
have been characterized. Melt is a protein that reversibly clusters and
translocates to the membrane in response to small temperature changes, enabling
flexible regulation of cellular processes in vitro and in vivo.",
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the human body beyond its role in oxygen and nutrient transport. We explored
vascular cell heterogeneity by integrating single-cell RNA-sequencing data from
the Human Cell Atlas and ongoing collaborations across 19 healthy human organs
and tissues, and highlight shared and distinct molecular features of vascular
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fratricide showed encouraging rates of clinical response and persistence of
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treated with the Trop-2-directed antibody\u2013drug conjugate sacituzumab
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prolonged progression-free survival compared with patients treated with
chemotherapy.",
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Healthy Longevity Initiative, this analysis finds that the economic impact of
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research priorities for the field of GLP-1 receptor agonists such as
semaglutide, from sex differences to how they might treat addiction.",
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with specific clinical trajectories",
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metabolic dysfunction-associated steatotic liver disease (MASLD) characterized
by distinct clinical trajectories. The two types of MASLD show similar liver
histology, but one is liver-specific with low cardiovascular risk while the
other has high cardiovascular risk and high metabolic risk.",
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checkpoint molecule CTLA-4 after stimulation in a microbiota-dependent manner,
and advances to support CTLA-4<sup>+</sup> ILC3s may represent a treatment
opportunity in IL-23-driven chronic inflammation.",
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inflatable air sac structure between the major flight muscles\u2014in 68
avian species reveals that the respiratory system has a role in the mechanics of
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are interpreted as having been left by a crown-group sponge fossil,
<i>Helicolocellus cantori</i> gen. et sp. nov., characterized by an organic
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genome enables identification of host factors and provides insights into
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amino acid sequences present in proteins with equally small segments of the
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African, Amazonian and Southeast Asian tropical forests suggests that, despite
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~2.2% of tropical tree species in each region, constitute half of Earth\u2019s
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        "title": "Disease-specific tau filaments assemble via polymorphic intermediates",
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appears in a recently synthesized antiferromagnet showing a strong
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stable transcription of a single allele.",
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Mesolithic, Neolithic and Early Bronze Age periods show that two major
population turnovers occurred over just 1,000 years in Neolithic Denmark,
resulting in dramatic changes in the genes, diet and physical appearance of the
local people, as well as the landscape in which they lived.",
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membrane-bound receptor Ror2 are transported between cells via cellular
protrusions called cytonemes to initiate paracrine Wnt5b signalling in cells
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found that school performance was a stronger predictor of young adult death than
parental education or earnings and largely explained the socioeconomic gradient
in early mortality.",
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targets and pathways involved in problematic alcohol use, which highlighted
potential therapeutic targets and emphasized the need for tailored, gene-
specific treatment strategies for effective intervention and the prevention of
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experienced participants used a elicitation protocol to make better-than-chance
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of family environment, childhood educational and occupational aspirations, and
links to health.",
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        "title": "Social and genetic diversity in first farmers of central
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(5500\u20135000 <span>bce</span>). They find differentiated genetic networks but
no evidence of unequal access to resources linked to sex or kin.",
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can forecast the collapse of scientific \u2018bubbles\u2019, highlighting that
sustained scientific advancement requires diverse audiences.",
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diffusion of biomedical knowledge in the literature, we find that scientific
ideas experience popularity booms and busts when knowledge diffusion is
constricted, which leads to adverse consequences for science and scientists. Our
work highlights the need for research to have an effect on diverse audiences to
achieve sustained scientific advancement.",
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labelling policy in England was associated with a change in calories consumed in
the out-of-home food sector. Our findings suggest that the introduction of the
policy was not associated with a significant decrease in calories purchased or
consumed.",
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        "url": "/articles/s41562-024-02046-9",
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that perceptions of scientists\u2019 intellectual humility positively affect the
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        "url": "/articles/s41562-024-02048-7",
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across 23 countries and find that Internet use is associated with reduced
depressive symptoms, higher life satisfaction and better self-reported health.",
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election seasons of 2017\u20132020 were forwarded without the sharer first
clicking on them. Extreme and user-aligned political content was shared more,
aiding the spread of partisan misinformation.",
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literacy goals for pupils in low- and middle-income countries",
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that underpin reading comprehension.",
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        "url": "/articles/s41562-024-02040-1",
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        "url": "/articles/s41562-024-02038-9",
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        "url": "/articles/s41562-024-02047-8",
        "description": "Working memory is constructive in nature. In three electroencephalography/magnetoencephalography studies, Fan et al. provide converging behavioural and neural evidence that a one-dimensional syllable sequence is automatically reorganized into a two-dimensional representation in working memory, which reflects the hierarchical organization of language.",
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        "title": "Large-scale exome sequencing identified 18 novel genes for neuroticism in 394,005 UK-based individuals",
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        "title": "Web-browsing patterns reflect and shape mood and mental health",
        "authors": [
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        "type": "Research"
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        "title": "Ethics of genomic research on occupational status",
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        "url": "/articles/s41562-024-02082-5",
        "description": "Genomic studies of social outcomes raise ethical considerations that heighten researcher obligations to responsibly conduct and communicate their work. A study by Akimova et al. finds that most intergenerational transmission of occupational status can be ascribed to nongenetic factors, and raises questions of how such knowledge should be used and how it might be misused.",
        "type": "News &amp; Views"
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        "title": "Becoming the ideal woman-of-colour academic for everyone but me",
        "authors": [
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        "type": "Comments &amp; Opinion"
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        "title": "Intergenerational persistence of poverty in five high-income countries",
        "authors": [
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        "url": "/articles/s41562-024-02029-w",
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        "description": "Children born in poverty are more likely to experience poverty as adults, but this likelihood is greater in the United States than in Australia, Denmark, Germany or the United Kingdom. The authors examine what mediates the differences in intergenerational poverty between countries.",
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        "title": "Effectiveness of ex ante honesty oaths in reducing dishonesty depends on content",
        "authors": [
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        "title": "Child literacy in low- and middle-income countries",
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        "description": "Despite rising school enrolments, in many places learning achievement has remained very low. Crawford et al. analyse data on half a million pupils from 48 low- and middle-income countries and find that children fall further behind literacy benchmarks with each grade. They suggest that a greater focus on phonics instruction and decoding skills may improve outcomes.",
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        "authors": [
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        "title": "Internet use and mental wellbeing in older adults",
        "authors": [
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        "description": "Mental health disorders among older adults rose during the COVID-19 pandemic, and have since continued to increase. A recent study provides timely insights into a positive association between internet use and mental wellbeing among adults aged 50 and older across 23 countries, which highlights the potential of digital technology for addressing these challenges.",
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        "title": "Web browsing reflects and shapes mood",
        "authors": [
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        "description": "Mental health problems pose a great burden for
today\u2019s societies. The internet and social media are blamed for
contributing to the burden, but the evidence to date is inconsistent. Kelly and
Sharot shed light on the causal relationships while focusing on the affective
properties of website content, and show how web browsing reflects and shapes
mental health.",
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        "title": "Science and Hinduism share the vision of a quest for truth",
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universe and is not antithetical to science. Rohitash Chandra explains why he
sees value in bringing together science and spirituality in the quest for
knowledge.",
        "type": "Comments &amp; Opinion"
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        "title": "Racial biases in polygraphs and their legal implications",
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        "url": "/articles/s41562-024-02025-0",
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        "title": "Intergenerational poverty persistence",
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        "description": "Intergenerational mobility \u2014 an adult\u2019s
ability to exit poverty and earn more than their parents \u2014 is falling in
the USA. Parolin et al. compare intergenerational poverty persistence (a measure
of immobility) in the USA to four peer countries and conclude that
disproportionately high poverty persistence in the USA is due to a weak safety
net.",
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        "description": "Honesty oaths are commonly used to promote ethical behaviour, but their effectiveness is not well understood. A mega-study involving thousands of people shows that taking an oath to be honest can reduce tax evasion in an online economic game.",
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        "title": "The mental health effects of the tenure track system in China",
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        "title": "The US media should rethink coverage of firearm violence",
        "authors": [
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        "title": "How to evaluate the cognitive abilities of LLMs",
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        "title": "How trait impressions of faces shape subsequent mental state inferences",
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        "description": "In this Stage 2 Registered Report, Lin et al. show that trait impressions of faces are correlated with and causally influence mental state inferences from the same faces.",
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        "title": "Report uncertainty information to improve trust in science",
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        "url": "/articles/s41562-024-02084-3",
        "description": "The results of scientific studies should be accompanied by information that individuals can use to make uncertainty judgements. By including this information, we might increase trust in the scientific process. We advocate for scientists to use quantitative uncertainty informative data to provide this information when reporting results.",
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        "title": "Incorporate climate injustice into carbon labels",
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        "title": "How COVID-19 has changed tourists\u2019 behaviour",
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methodological and practical implications, but also entails addressing
challenges related to internal and external validity, replicability and ethics.
Here we illuminate the issues involved in each step of the research process and
offer practical recommendations to address them.",
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        "title": "Human culture is uniquely open-ended rather than uniquely
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        "url": "/articles/s41562-024-02035-y",
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Morgan and Feldman re-examine existing theoretical accounts and propose that,
contrary to previous belief, cumulative change and high transmission fidelity
are not unique to human culture.",
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        "url": "/articles/s41562-024-02006-3",
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intervention, reduced prison adjudications by 50%. Following release, social
bonding was related to desistance and community attitudes, highlighting the need
to connect incarcerated people with law-abiding groups.",
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sense. Using a precise sniff-triggered apparatus, Wu et al. show that people can
distinguish fine odour dynamics, with a temporal sensitivity within 120
milliseconds\u2014or well within a single sniff.",
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in Argentina more than a one-way reminder",
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more than tripled COVID-19 vaccination uptake in Chaco province, Argentina, and
nearly doubled it compared with a one-way message reminder.",
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conditions: a systematic review and meta-analysis",
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        "authors": [
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        "description": "Intrusive memories are a hallmark symptom of post-traumatic stress disorder (PTSD). This preregistered meta-analysis investigates how experimental techniques alter intrusive memory frequency, intrusion-related distress and symptoms arising from lab-analogue trauma exposure.",
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        "title": "Causal effect of video gaming on mental well-being in Japan 2020\u20132022",
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        "url": "/articles/s41562-024-01948-y",
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        "title": "Neural populations in the language network differ in the size of their temporal receptive windows",
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        "url": "/articles/s41562-024-01944-2",
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        "title": "Consonant lengthening marks the beginning of words across a diverse sample of languages",
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        "description": "Blum et al. report evidence of lengthening of word-initial consonants across a diverse sample of 51 languages. On average, these consonants are 13\u2009ms longer than word-medial ones, helping mark word boundaries in continuous speech, which is crucial for understanding speech.",
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        "title": "State-level anti-transgender laws increase past-year suicide attempts among transgender and non-binary young people in the USA",
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        "title": "Observational and genetic evidence disagree on the association between loneliness and risk of multiple diseases",
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        "title": "Replay shapes abstract cognitive maps for efficient social navigation",
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        "url": "/articles/s41562-024-01990-w",
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        "title": "(Re)Emerging disease and conflict risk in Africa, 1997\u20132019",
        "authors": [
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        "url": "/articles/s41562-024-01929-1",
        "description": "Koren and Bukari use geolocated data on zoonotic diseases in Africa (1997\u20132019) to examine their impact on conflict. They find that zoonotic diseases are linked to a decrease in state-initiated civil wars but an increase in social conflicts involving identity militias.",
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        "title": "Fact-checker warning labels are effective even for those who distrust fact-checkers",
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        "url": "/articles/s41562-024-01973-x",
        "description": "Do fact-checker warning labels work for those who distrust fact-checkers? Martel and Rand analyse 21 experiments and find that warning labels reduce belief in, and sharing of, false posts both on average and for those highly distrusting of fact-checkers.",
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        "title": "Distributed representations of behaviour-derived object dimensions in the human visual system",
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        "url": "/articles/s41562-024-01977-7",
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minority (URM) faculty in the USA face barriers in the promotion and tenure
process, receiving more negative votes and fewer unanimous positive decisions at
the college level. This is partly due to a double standard: URM faculty are held
to a higher standard than non-URM faculty in terms of scholarly productivity.",
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        "title": "Maternal adiposity and perinatal and offspring outcomes: an
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        "url": "/articles/s41562-024-01994-6",
        "description": "This umbrella review of 194 observational meta-analyses
and 748 interventional studies finds that maternal adiposity is associated with
12 adverse perinatal and offspring outcomes. Maternal weight loss interventions
can only reduce some of these effects.",
        "type": "Research"
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        "title": "A systematic review and network meta-analysis of population-
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        "url": "/articles/s41562-024-02002-7",
        "description": "In this systematic review and meta-analysis of
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measures for curbing smoking behaviour.",
        "type": "Research"
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        "title": "A 27-country test of communicating the scientific consensus on climate change",
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        "url": "/articles/s41562-024-01928-2",
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        "type": "Research"
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        "title": "The genetic landscape of neuro-related proteins in human plasma",
        "authors": [
          "Linda Repetto",
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        "url": "/articles/s41562-024-01963-z",
        "description": "Repetto et al. provide an analysis of the genetic basis of variation of neuro-related protein levels in plasma and link this to human behaviour and disorders.",
        "type": "Research"
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        "title": "Conceptual structure and the growth of scientific knowledge",
        "authors": [
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        "url": "/articles/s41562-024-01957-x",
        "description": "Analysing 2,605,224 papers spanning five decades from both the social sciences and the physical sciences, the authors examine how scientific knowledge grows. Their findings suggest that while the organization of scientific concepts is important for the growth of knowledge, the mechanisms vary across time.",
        "type": "Research"
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        "title": "A large-scale field experiment on participatory decision-making in China",
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        "url": "/articles/s41562-024-01964-y",
        "description": "Wu et al. show that involving citizens in local decision-making (participatory budgeting) improves civic engagement in a Chinese context.",
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        "title": "Public opinion, racial bias and labour market outcomes in the
USA",
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        "url": "/articles/s41562-024-01904-w",
        "description": "Majlesi et al. show that Asian workers in occupations
with a higher likelihood of face-to-face interactions were more likely to become
unemployed after the start of the COVID-19 pandemic due to shifts in public
opinion. People who voted for Donald Trump in 2016 had more negative shifts in
perceived favourability of Asians.",
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        "title": "Theta phase precession supports memory formation and retrieval
of naturalistic experience in humans",
        "authors": [
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        "url": "/articles/s41562-024-01983-9",
        "description": "Zheng et al. show neural evidence for a functional role
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        "type": "Research"
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        "title": "Moving beyond processing- and analysis-related variation in
resting-state functional brain imaging",
        "authors": [
          "Xinhui Li",
          "Nathalia Bianchini Esper",
          "Michael P. Milham"
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        "url": "/articles/s41562-024-01942-4",
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across different functional magnetic resonance imaging preprocessing pipelines.
Due to these variations, using seemingly similar minimal preprocessing does not
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for various everyday and innovation-related problems, compared with not using
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robust regardless of whether the problem required consideration of many (versus
few) constraints and whether the problem was viewed as requiring empathetic
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She discusses how STEM sign lexicon development contributes to inclusive
education and which challenges still need to be overcome.",
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        "title": "Balancing autonomy and expertise in autonomous synthesis laboratories",
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environment that uses finite elements for shape optimization in soft matter.
Given an energy functional that incorporates physical boundaries and effects
such as elasticity and electromagnetism, together with additional constraints to
be satisfied, Morpho predicts the optimized shape and structure adopted by the
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electronic structures, called MEHnet, is developed to predict various molecular
properties in a unified framework, approaching chemical accuracy while
exhibiting local DFT-level computational costs.",
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        "description": "An extensive audit of large language models reveals that
numerous models mirror the \u2018us versus them\u2019 thinking seen in human
behavior. These social prejudices are likely captured from the biased contents
of the training data.",
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learning framework that uses the stochastic properties of memristors for in situ
probabilistic computations. This framework can greatly improve the efficiency
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robot skill-learning task.",
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and artificial systems.",
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randomized networks that preserve the weighted degree sequence. The procedure
outperforms standard rewiring algorithms and extends to multiple network types,
including directed and signed networks.",
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data structure that encodes 200,000 human genomes in just
5\u201326\u2009gigabytes per chromosome. Computation on GRG via graph traversal
greatly accelerates genome-wide analysis.",
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social identity biases similar to humans, having favoritism toward ingroups and
hostility toward outgroups. These biases persist across models, training data
and real-world human\u2013LLM conversations.",
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method that leverages machine learning and atomistic simulations to predict the
most likely crystallization products upon annealing of amorphous precursors. The
a<sup>2</sup>c tool was demonstrated on a variety of materials, including
oxides, nitrides and metallic glasses, and can assist researchers in discovering
synthesis pathways for materials design.",
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to learn geometry-dependent solution operators of partial differential equations
(PDEs). The framework enables scalable and fast approximations of PDE solutions
on a variety of 3D geometries.",
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Schr\u00f6dinger equation with a deep neural network",
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        "url": "/articles/s43588-024-00730-4",
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erroneous annotations and ambiguous cell states, infers trajectories from binary
labels and enables signal-aware analysis.",
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        "type": "Research"
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        "title": "A simulated <i>C. elegans</i> with biophysically detailed neurons and muscle dynamics",
        "authors": [],
        "url": "/articles/s43588-024-00740-2",
        "description": "We created an open-source model that simulates <i>Caenorhabditis elegans</i> in a closed-loop system, by integrating simulations of its brain, its physical body, and its environment. BAAIWorm replicated <i>C. elegans</i> locomotive behaviors, and synthetic perturbations of synaptic connections impacted neural control of movement and affected the embodied motor behavior.",
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        "title": "On the path toward brain-scale simulations",
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        "description": "We continue our yearly tradition of highlighting our favorite cover suggestions that did not make it as covers at <i>Nature Computational Science</i>.",
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        "title": "Improving the prediction of protein stability changes upon mutations by geometric learning and a pre-training strategy",
        "authors": [
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        "title": "Deep learning training dynamics analysis for single-cell data",
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        "description": "Inspired by recent approaches for natural language processing and computer vision, we developed Annotatability, a framework that analyzes deep neural network training dynamics to interpret pre-annotated single-cell and spatial omics data. Annotatability identified erroneous annotations and ambiguous cell states, inferred trajectories from binary labels, and revealed underlying biological signals.",
        "type": "News &amp; Views"
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        "title": "Teaching spin symmetry while learning neural network wave functions",
        "authors": [
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        "title": "Reliable deep learning in anomalous diffusion against out-of-distribution dynamics",
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        "authors": [
          "Jun-Tian Ye",
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        "url": "/articles/s43588-024-00722-4",
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        "authors": [
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        "url": "/articles/s43588-024-00698-1",
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        "title": "Harnessing the power of DNA for computing",
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        "title": "MassiveFold: unveiling AlphaFold\u2019s hidden potential with optimized and parallelized massive sampling",
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        "type": "Research"
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        "title": "The motive cocktail in altruistic behaviors",
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        "url": "/articles/s43588-024-00685-6",
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        "title": "Harnessing deep learning to build optimized ligands",
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        "url": "/articles/s43588-024-00725-1",
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        "type": "News &amp; Views"
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        "title": "Enhancing protein stability prediction with geometric learning
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predict protein fitness and stability after the introduction of mutations. The
work focuses on combining different data and pre-training to overcome data
scarcity.",
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        "title": "Extracting reliable quantum outputs for noisy devices",
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        "url": "/articles/s43588-024-00713-5",
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noise-free quantum expectation values from noisy samples, promising potential
applications in quantum optimization and machine learning.",
        "type": "News &amp; Views"
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        "title": "E-waste challenges of generative artificial intelligence",
        "authors": [
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        "url": "/articles/s43588-024-00712-6",
        "description": "Generative artificial intelligence (GAI) is driving a
surge in e-waste due to intensive computational infrastructure needs. This study
emphasizes the necessity for proactive implementation of circular economy
practices throughout GAI value chains.",
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        "title": "Deep learning large-scale drug discovery and repurposing",
        "authors": [
          "Min Yu",
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profiling changes in mitochondrial phenotypes, allowing for the identification
of various drugs\u2019 mechanism of action.",
        "type": "Research"
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        "url": "/articles/s43588-024-00674-9",
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        "description": "Active machine learning is employed in academia and industry to support drug discovery. A recent study unravels the factors that influence a deep learning models\u2019 ability to guide iterative discovery.",
        "type": "News &amp; Views"
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        "title": "Automated discovery of symbolic laws governing skill acquisition from naturally occurring data",
        "authors": [
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        "url": "/articles/s43588-024-00629-0",
        "description": "This paper introduces an algorithm to uncover laws of skill acquisition from naturally occurring data. By combining deep learning and symbolic regression, it accurately identifies cognitive states and extracts algebraic equations.",
        "type": "Research"
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        "title": "The decomposition of perturbation modeling",
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        "url": "/articles/s43588-024-00706-4",
        "description": "A recent study proposes a strategy for the prediction of genetic perturbation outcomes by breaking it down into three subtasks: identifying differentially expressed genes, determining expression change directions, and estimating gene expression magnitudes.",
        "type": "News &amp; Views"
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        "title": "An integrative data-driven model simulating <i>C. elegans</i> brain, body and environment interactions",
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        "url": "/articles/s43588-024-00738-w",
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        "description": "BAAIWorm is an integrative data-driven model of <i>C. elegans</i> that simulates interactions between the brain, body and environment. The biophysically detailed neuronal model is capable of replicating the zigzag movement observed in this species.",
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        "title": "Simulation and assimilation of the digital human brain",
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        "title": "Automated customization of large-scale spiking network models to neuronal population activity",
        "authors": [
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        "url": "/articles/s43588-024-00688-3",
        "description": "An automatic framework, SNOPS, is developed for configuring a spiking network model to reproduce neuronal recordings. It is used to discover previously unknown limitations of spiking network models, thereby guiding model development.",
        "type": "Research"
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        "title": "Joint inference of discrete cell types and continuous type-specific variability in single-cell datasets with MMIDAS",
        "authors": [
          "Yeganeh Marghi",
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        "url": "/articles/s43588-024-00683-8",
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